



Exercise-1

Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

CHEMICAL KINETICS-I

Section (A) : Rate of chemical reaction and Dependence of Rate : Basic

A-1. The oxidation of iodide ion by peroxy disulphate ion is described by the equation :



(a) If $-\frac{\Delta[\text{S}_2\text{O}_8^{2-}]}{\Delta t} = 1.5 \times 10^{-3} \text{ Ms}^{-1}$ for a particular time interval, what is the value of $-\frac{\Delta[\text{I}^-]}{\Delta t}$ for the same time interval ?

(b) What is the average rate of formation of SO_4^{2-} during that time interval ?

A-2. In the following reaction $2\text{H}_2\text{O}_2 (\text{aq}) \longrightarrow 2\text{H}_2\text{O} (\ell) + \text{O}_2 (\text{g})$ rate of formation of O_2 is 36 g min^{-1} ,

(a) What is rate of formation of H_2O ?

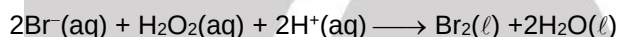
(b) What is rate of disappearance of H_2O_2 ?

A-3. Hydrogenation of vegetable ghee at 25°C reduces the pressure of H_2 from 2 atm to 1.2 atm in 50 minute. Calculate the rate of reaction in terms of change :

(a) In pressure per minute.

(b) In molarity per second.

A-4. The stoichiometric equation for the oxidation of bromide ions by hydrogen peroxide in acid solution is



Since the reaction does not occur in one stage, the rate equation does not correspond to this stoichiometric equation but is $\text{rate} = k[\text{H}_2\text{O}_2][\text{H}^+][\text{Br}^-]$.

(a) If the concentration of H_2O_2 is increased by a factor of 3, by what factor is the rate of consumption of Br^- ions increased.

(b) If, under certain conditions, the rate of consumption of Br^- ions is $7.2 \times 10^{-3} \text{ mole dm}^{-3} \text{ s}^{-1}$, what is the rate of consumption of hydrogen peroxide. What is the rate of production of bromine.

(c) What is the effect on the rate constant k of increasing the concentration of bromide ions.

(d) If by the addition of water to the reaction mixture the total volume were doubled, what would be the effect on the rate of change of the concentration of Br^- . What would be the effect on the rate constant k .

Section (B) : Integrated rate law : Zero and First Order Reaction

B-1. For the zero order reaction $\text{A} \longrightarrow 2\text{B}$, the rate constant is $2 \times 10^{-6} \text{ M min}^{-1}$. The reaction is started with 10 M of A.

(i) What will be the concentration of A after 2 days? (ii) What is the initial half-life of the reaction ?

(iii) In what time, the reaction will complete?

B-2. Substance A reacts according to a first order rate law with $k = 5 \times 10^{-5} \text{ s}^{-1}$.

(a) If the initial concentration of A is 1.0 M, what is initial rate .

(b) What is the rate after 1.0 hour. (Given $e^{-0.18} = 0.84$)



B-3. A and B are two different chemical species undergoing 1st order decomposition with half lives equal to 5 sec. and 7.5 sec. respectively. If the initial concentration of A and B are in the ratio 3 : 2. Calculate $\frac{C_{A_t}}{C_{B_t}}$ after three half lives of A. Report your answer after multiplying it with 100.

B-4. In the biological processes, the time taken by certain virus or bacteria to double its population, is called generation time. In milk, at 37°C, lactobacillus acidophilus has a generation time of about 75 min. Calculate the population relative to the initial value at 60 min.

Section (C) : Integrated Rate law : Second Order & Pseudo first order reaction

- C-1.** The rate constant for a certain second order reaction is $8 \times 10^{-5} \text{ M}^{-1} \text{ min}^{-1}$. How long will it take a 1 M solution to be reduced to 0.5 M in reactant? How long will it take from that point until the solution is 0.25 M in reactant.
- C-2.** Compound A and B react to form C and D in a reaction that was found to be second-order over all and second-order in A. The rate constant –at 30°C is $0.622 \text{ L mol}^{-1} \text{ min}^{-1}$. What is the half-life of A when $4.10 \times 10^{-2} \text{ M}$ of A is mixed with excess B?
- C-3.** The reaction $\text{CH}_3\text{CH}_2\text{NO}_2 + \text{OH}^- \longrightarrow \text{CH}_3\text{CH}^-\text{NO}_2 + \text{H}_2\text{O}$ obeys the rate law for pseudo first order kinetics in the presence of a large excess of hydroxide ion. If 1% of nitro ethane undergoes reaction in half a minute when the reactant concentration is 0.002 M. What is the pseudo first order rate constant? ($\log 99 = 1.996$)

Section (D) : Experimental method to calculate order and rate law of reaction and methods to monitor the progress of reaction

D-1. The following data are for the reaction $\text{A} + \text{B} \rightarrow \text{products}$:

conc. A (M)	conc. B (M)	Initial Rate ($\text{mol L}^{-1} \text{ s}^{-1}$)
0.1	0.1	4.0×10^{-4}
0.2	0.2	1.6×10^{-3}
0.5	0.1	2.0×10^{-3}
0.5	0.5	1.0×10^{-2}

- (i) What is the order with respect to A and B for the reaction?
- (ii) Calculate the rate constant.
- (iii) Determine the reaction rate when the concentrations of A and B are 0.20 M and 0.35 M, respectively.

D-2. (a) The half life period and initial concentration for a reaction are as follows. What is order of reaction?

Initial concentration	350	540	158
$t_{1/2}$	425	275	941

(b) The half-life period for the thermal decomposition of PH_3 at three different pressures are given below

Initial pressure (mm Hg)	707	79	37.5
Half-life (seconds)	84	84	83

What is the order of the reaction?



D-3. For the reaction $A \longrightarrow \text{products}$, the following data is given for a particular run.

time (min.) :	0	5	15	35
$\frac{1}{[A]} \text{ (M}^{-1}\text{)} :$	1	2	4	8

Determine the order of the reaction.

Section (E) : Method to monitor the progress of Reactions (Titration Method and Optical Activity Method)

E-1. For a first order reaction $A \longrightarrow 2B + C$. It was found that at the end of 10 minutes from the start, the total optical rotation of the system was 60° and when the reaction is complete, it was 180° . The B and C are only optically active and initially only A was taken.

(i) What is the rate constant of the above reaction (in hour^{-1}) ?

(ii) At what time (in minute) from the start, total optical rotation will be 90° .

(Take $\log 2 = 0.3$, $\log 3 = 0.48$, $\log 7 = 0.85$, $\ln 10 = 2.3$)

E-2. The kinetic of hydrolysis of methyl acetate in excess dilute HCl at 25°C were followed by withdrawing 2 mL of the reaction mixture at intervals of (t), adding 50 mL water and titrating with baryta water. Determine the velocity constant of hydrolysis.

t (in minute)	0	75	119	259	∞
Titre value (in mL)	19.24	24.20	26.60	32.23	42.03
$[\ln \frac{22.79}{17.83} = 0.2454, \ln \frac{22.79}{15.43} = 0.39, \ln \frac{22.79}{9.8} = 0.8439]$					

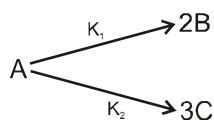
E-3. The decomposition of hydrogen peroxide in an aqueous solution is a first order reaction. It can be studied by titrating quickly 10 mL portions of reactions mixture at various times from the $t = 0$ of reaction against a standard solution of KMnO_4 . Volume of KMnO_4 solution used in each case is proportional to the remaining concentration of H_2O_2 .

From the following data calculate the rate constant of the reaction,

Time (seconds)	0	600	1200
KMnO_4 solution used (mL)	22.8	13.8	8.2

Section (F) : Complication in 1st Order Reaction

F-1. For a first order reaction



Percentage of B in the product is 30%. Calculate the value of k_1 and k_2 . ($(t_{1/2})_{\text{over all}} = 100 \text{ hr}$)

F-2. An organic compound dissociates into n parallel first order reactions simultaneously and produces n different products $P_1, P_2, P_3, \dots, P_n$ having rate constants $k, 2k, 3k, \dots, nk$ and activation energies $E, 2E, 3E, \dots, nE$ respectively. Calculate the overall energy of activation of the compound P.



CHEMICAL KINETICS-II

Section (G) : Temperature dependence of rate

G-1. From the following data for the reaction between A and B.

	[A] (mole dm ⁻³)	[B] (mole dm ⁻³)	Initial rate (mole dm ⁻³ min ⁻¹)	
			300 K	320 K
(i)	2.5×10^{-4}	3.0×10^{-5}	5.0×10^{-4}	2.0×10^{-3}
(ii)	5.0×10^{-4}	6.0×10^{-5}	4.0×10^{-3}	
(iii)	1.0×10^{-3}	6.0×10^{-5}	1.6×10^{-2}	

Calculate the following

(i) The order w.r.t. A and w.r.t. B

(iii) The energy of activation

$[e^{-22.18} = 2.3283 \times 10^{-10}, \ln 2 = 0.693]$

(ii) The rate constant at 300 K

(iv) The pre-exponential factor

G-2. Find out the percentage of the reactant molecules crossing over the activation energy barrier at 325 K, given that $\Delta H_{325} = 0.12$ kcal, $E_{a(b)} = +0.02$ kcal.

G-3. Consider the following figure for the reaction :



Answer the following :

(A) Energy level of (A + B)

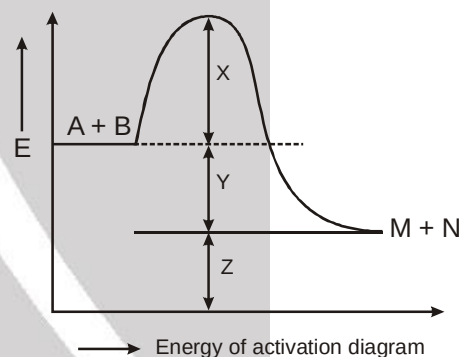
(B) Energy level of (M + N)

(C) Threshold energy level for the reaction

(D) Energy of activation for the forward reaction.

(E) Energy of activation for the backward reaction.

(F) ΔH or heat of reaction.

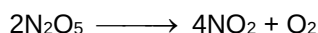
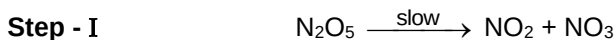


→ Energy of activation diagram

Section (H) : Catalyst dependence of rate and type of Reaction and Determination of rate law and order with the help of given mechanism

H-1. On introducing a catalyst at 500 K, the rate of a first order increase to 1.718 times. The activation energy in the presence of a catalyst is 6.059 KJ/mole. The slope of the plot of $\ln k$ (sec⁻¹) against $1/T$ in the absence of catalyst is $(R = 8.3$ J/K mol, $\ln 1.718 = 0.54)$.

H-2. The thermal decomposition of N_2O_5 occurs in the following steps.



suggest the rate expression.

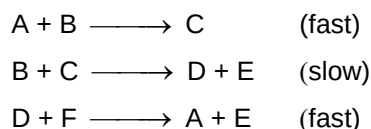


H-3. The reaction $2\text{NO}(\text{g}) + \text{Br}_2(\text{g}) \longrightarrow 2\text{NOBr}(\text{g})$, obeys the following mechanism.

- (i) $\text{NO}(\text{g}) + \text{Br}_2(\text{g}) \xrightleftharpoons{\text{Fast}} \text{NOBr}_2(\text{g})$
 (ii) $\text{NOBr}_2(\text{g}) + \text{NO}(\text{g}) \xrightarrow{\text{Slow}} 2\text{NOBr}(\text{g})$

Suggest the rate expression.

H-4. Given the following steps in the mechanism for a chemical reaction :



At any time $[\text{C}]$ is directly proportional to $[\text{A}]$.

- (a) What is the stoichiometric equation for the reaction ?
 (b) Which species, if any, are catalysts in this reaction ?
 (c) Which species, if any, are intermediates in this reaction?
 (d) Write the rate law for the rate-determining step.
 (e) Write the rate law for this reaction.
 (f) What is overall order of the reaction ?

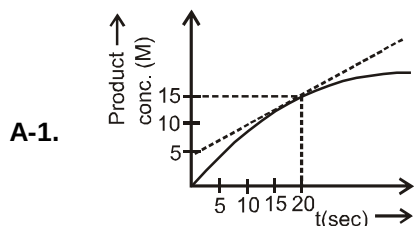
Section (I) : Radio Activity

- I-1. The activity of a radio-isotope falls to 12.5% in 90 days. Compute the half life and decay constant of the radio-isotope.
 I-2. Radioactive decay is a first order process. Radioactive carbon in wood sample decays with a half life of 5770 years. What is the rate constant (in years^{-1}) for the decay? What fraction would remain after 11540 years.
 I-3. 1 gram of ${}_{79}\text{Au}^{198}$ ($t_{1/2} = 65$ hrs) decays by beta emission and produces stable mercury. How much mercury will be present after 260 hours.
 I-4. A sample of U^{238} (half life = 4.5×10^9 year) ore is found to contain 23.8 g of U^{238} and 20.6 g of Pb^{206} . Calculate the age of the ore.

PART - II : ONLY ONE OPTION CORRECT TYPE

CHEMICAL KINETICS-I

Section (A) : Rate of chemical reaction and Dependence of Rate : Basic



Rate of formation of product at $t = 20$ seconds is

- (A) 0.5 M S^{-1} (B) 1 M S^{-1} (C) 1.5 M S^{-1} (D) 2 M S^{-1}



A-2. In the following reaction : $x\text{A} \longrightarrow y\text{B}$

$$\log \left[-\frac{d[\text{A}]}{dt} \right] = \log \left[\frac{d[\text{B}]}{dt} \right] + 0.3$$

where –ve sign indicates rate of disappearance of the reactant. Thus, x : y is :

- (A) 1 : 2 (B) 2 : 1 (C) 3 : 1 (D) 3 : 10

A-3. Rate of formation of SO_3 in the following reaction $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$ is 100 g min^{-1} . Hence rate of disappearance of O_2 is :

- (A) 50 g min^{-1} (B) 40 g min^{-1} (C) 200 g min^{-1} (D) 20 g min^{-1}

A-4. $a\text{A} + b\text{B} \longrightarrow \text{Product}$, $\frac{dx}{dt} = k [\text{A}]^a [\text{B}]^b$. If concentration of A is doubled, rate is four times. If concentration of B is made four times, rate is doubled. What is relation between rate of disappearance of A and that of B ?

- (A) $-\{d[\text{A}] / dt\} = -\{d[\text{B}] / dt\}$ (B) $-\{d[\text{A}] / dt\} = -\{4 d[\text{B}] / dt\}$
(C) $-\{4 d[\text{A}] / dt\} = -\{d[\text{B}] / dt\}$ (D) None of these

A-5. For the reaction, $2\text{NO}(\text{g}) + 2\text{H}_2(\text{g}) \longrightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ the rate expression can be written in the following ways:

$$\{d[\text{N}_2] / dt\} = k_1 [\text{NO}][\text{H}_2] ; \{d[\text{H}_2\text{O}] / dt\} = k[\text{NO}][\text{H}_2] ; \{-d[\text{NO}] / dt\} = k'_1 [\text{NO}] [\text{H}_2] ; \{-d[\text{H}_2] / dt\} = k''_1 [\text{NO}][\text{H}_2]$$

The relationship between k, k_1 , k'_1 and k''_1 is :

- (A) $k = k_1 = k'_1 = k''_1$ (B) $k = 2k_1 = k'_1 = k''_1$ (C) $k = 2k'_1 = k_1 = k''_1$ (D) $k = k_1 = k'_1 = 2 k''_1$

A-6. For a reaction $p\text{A} + q\text{B} \rightarrow \text{products}$, the rate law expression is $r = k[\text{A}]^1 [\text{B}]^m$, then :

- (A) $(p+1) < (1+m)$ (B) $(p+q) > (1+m)$
(C) $(p+q)$ may or may not be equal to $(1+m)$ (D) $(p+q) = (1+m)$

A-7. For the irreversible process, $\text{A} + \text{B} \longrightarrow \text{products}$, the rate is first-order w.r.t. A and second-order w.r.t. B. If 1.0 mol each of A and B introduced into a 1.0 L vessel, and the initial rate was $1.0 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$, rate when half reactants have been turned into products is :

- (A) $1.25 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$ (B) $1.0 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$
(C) $2.50 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$ (D) $2.0 \times 10^{-2} \text{ mol L}^{-1} \text{ s}^{-1}$

A-8. If rate constant is numerically the same for the three reactions of first, second and third order respectively. Assume all the reactions of the kind $\text{A} \rightarrow \text{products}$. Which of the following is correct :

- (A) if $[\text{A}] = 1$ then $r_1 = r_2 = r_3$ (B) if $[\text{A}] < 1$ then $r_1 > r_2 > r_3$
(C) if $[\text{A}] > 1$ then $r_3 > r_2 > r_1$ (D) All

Section (B) : Integrated rate law : Zero and First Order Reaction

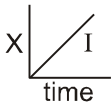
B-1. The rate constant of the reaction $\text{A} \rightarrow 2\text{B}$ is $1.0 \times 10^{-3} \text{ mol lit}^{-1} \text{ min}^{-1}$, if the initial concentration of A is $1.0 \text{ mole lit}^{-1}$ what would be the concentration of B after 100 minutes.

- (A) 0.1 mol lit^{-1} (B) 0.2 mol lit^{-1} (C) 0.9 mol lit^{-1} (D) 1.8 mol lit^{-1}

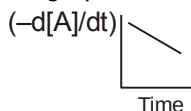
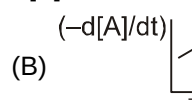
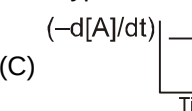
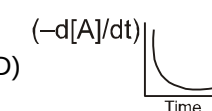
B-2. A drop of solution (volume 0.05 mL) contains 3.0×10^{-6} moles of H^+ . If the rate constant of disappearance of H^+ is $1.0 \times 10^7 \text{ mole litre}^{-1} \text{ sec}^{-1}$. How long would it take for H^+ in drop to disappear :

- (A) $6 \times 10^{-8} \text{ sec}$ (B) $6 \times 10^{-7} \text{ sec}$ (C) $6 \times 10^{-9} \text{ sec}$ (D) $6 \times 10^{-10} \text{ sec}$

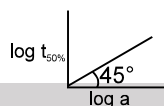


- B-3.** Graph between concentration of the product and time of the reaction $A \rightarrow B$ is of the type .

Hence graph between $-d[A]/dt$ and time will be of the type :

- (A)  (B)  (C)  (D) 

- B-4.** What will be the order of reaction and rate constant for a chemical change having $\log t_{50\%}$ vs $\log$ concentration of (A) curves as :



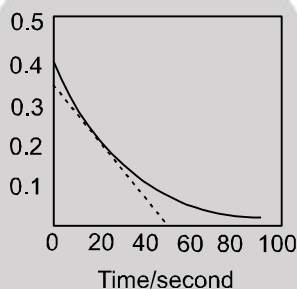
- (A) 0, $1/2$ (B) 1, 1 (C) 2, 2 (D) 3, 1

- B-5.** For a reaction $2A + B \rightarrow \text{product}$, rate law is $-\frac{d[A]}{dt} = k[A]$. At a time when $t = \frac{1}{k}$, concentration of the reactant is : (C_0 = initial concentration)

- (A) $\frac{C_0}{e}$ (B) $C_0 e$ (C) $\frac{C_0}{e^2}$ (D) $\frac{1}{C_0}$

- B-6.** Two substances A ($t_{1/2} = 5$ min) and B ($t_{1/2} = 15$ min) are taken in such a way that initially $[A] = 4[B]$. The time after which both the concentration will be equal is : (Assume that reaction is first order)
- (A) 5 min (B) 15 min (C) 20 min (D) concentration can never be equal

- B-7.** A reaction follows the given concentration (M)–time graph. The rate for this reaction at 20 seconds will be :



- (A) $4 \times 10^{-3} \text{ M s}^{-1}$ (B) $8 \times 10^{-2} \text{ M s}^{-1}$ (C) $2 \times 10^{-2} \text{ M s}^{-1}$ (D) $7 \times 10^{-3} \text{ M s}^{-1}$

- B-8.** In a first order reaction the reacting substance has half-life period of ten minutes. What fraction of the substance will be left after an hour the reaction has occurred ?

- (A) $1/6$ of initial concentration (B) $1/64$ of initial concentration
(C) $1/12$ of initial concentration (D) $1/32$ of initial concentration

- B-9.** A certain zero order reaction has $k = 0.025 \text{ M s}^{-1}$ for the disappearance of A. What will be the concentration of A after 15 seconds if the initial concentration is 0.50 M ?
- (A) 0.50 M (B) 0.375 M (C) 0.125 M (D) 0.060 M

Section (C) : Integrated Rate law : Second Order & Pseudo first order reaction

- C-1.** A reaction, which is second order, has a rate constant of $0.002 \text{ L mol}^{-1} \text{ s}^{-1}$. If the initial conc. of the reactant is 0.2 M . how long will it take for the concentration to become 0.0400 M ?
- (A) 1000 sec (B) 400 sec (C) 200 sec (D) 10, 000 sec



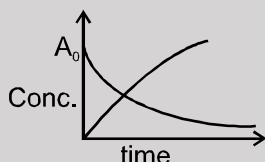
C-2. Which is not true for a second order reaction ?

- (A) It can have rate constant $1 \times 10^{-2} \text{ L mol}^{-1} \text{ s}^{-1}$
 (B) Its half-life is inversely proportional to its initial concentration
 (C) Time to complete 75% reaction is twice of half-life
 (D) $T_{50} = \frac{1}{k \times \text{Initial conc.}}$

C-3. For the reaction $2\text{NO}_2 \longrightarrow \text{N}_2\text{O}_2 + \text{O}_2$, rate expression is as follows :

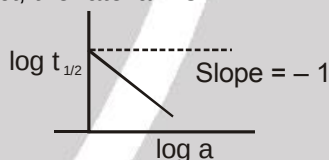
$-\frac{d[\text{NO}_2]}{dt} = k [\text{NO}_2]^n$, where $k = 3 \times 10^{-3} \text{ mol}^{-1} \text{ L sec}^{-1}$. If the rate of formation of oxygen is $1.5 \times 10^{-4} \text{ mol L}^{-1} \text{ sec}^{-1}$, then the molar concentration of NO_2 in mole L^{-1} is
 (A) 1.5×10^{-4} (B) 0.0151 (C) 0.214 (D) 0.316

C-4. At the point of intersection of the two curves shown, the conc. of B is given by.....for, $A \rightarrow nB$:



- (A) $\frac{nA_0}{2}$ (B) $\frac{A_0}{n-1}$ (C) $\frac{nA_0}{n+1}$ (D) $\left(\frac{n-1}{n+1}\right)A_0$

C-5. A graph between $\log t_{1/2}$ and $\log a$ (abscissa) a being the initial concentration of A in the reaction For reaction $A \rightarrow \text{Product}$, the rate law is



- (A) $\frac{-d[A]}{dt} = k$ (B) $\frac{-d[A]}{dt} = k [A]$ (C) $\frac{-d[A]}{dt} = k [A]^2$ (D) $\frac{-d[A]}{dt} = k [A]^3$

Section (D) : Experimental Method to calculate order and rate law of reaction and Methods to Monitor the progress of Reaction

D-1. The data for the reaction $A + B \rightarrow C$ is

Exp.	$[A]_0$	$[B]_0$	initial rate
1	0.012	0.035	0.10
2	0.024	0.035	0.80
3	0.012	0.070	0.10
4	0.024	0.070	0.80

The rate law is

- (A) $r = k [B]^3$ (B) $r = k [A]^3$ (C) $r = k [A] [B]^4$ (D) $r = k [A]^2 [B]^2$

D-2. The kinetic data for the given reaction $A(g) + 2B(g) \longrightarrow C(g)$ is provided in the following table for three experiments at 300 K.

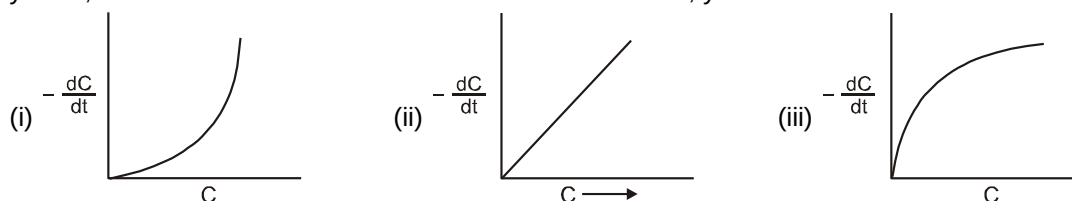
Ex. No.	$[A/M]$	$[B/M]$	[Initial rate (M sec^{-1})]
1	0.01	0.01	6.930×10^{-6}
2	0.02	0.01	1.386×10^{-5}
3	0.02	0.02	1.386×10^{-5}

In another experiment starting with initial concentration of 0.5 and 1 M respectively for A and B at 300 K. Find the rate of reaction after 50 minutes from start of experiment (in m/sec)?

- (A) 6.93×10^{-4} (B) 0.25×10^{-7} (C) 4.33×10^{-5} (D) 3.46×10^{-9}



D-3. In three different reactions, involving a single reactant in each case, a plot of rate of the reaction on the y-axis, versus concentration of the reactant on the x-axis, yields three different curves shown below.



What are the possible orders of the reactions (i), (ii), (iii).

- (A) 1, 2, 3 (B) 2, 1, 1/2 (C) 0, 1, 2 (D) 0, 1, 1/2

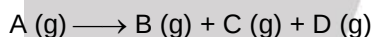
D-4. At 373 K, a gaseous reaction $A \rightarrow 2B + C$ is found to be of first order. Starting with pure A, the total pressure at the end of 10 min was 176 mm of Hg and after a long time when A was completely dissociated, it was 270 mm of Hg. The pressure of A at the end of 10 minutes was :

- (A) 94 mm of Hg (B) 47 mm of Hg (C) 43 mm of Hg (D) 90 mm of Hg

D-5. The reaction $A(s) \rightarrow 2B(g) + C(g)$ is first order. The pressure after 20 min and after very long time are 150 mm Hg and 225 mm Hg. The value of rate constant and pressure after 40 min are :

- (A) $0.05 \ln 1.5 \text{ min}^{-1}$, 200 mm (B) $0.5 \ln 2 \text{ min}^{-1}$, 300 mm
(C) $0.05 \ln 3 \text{ min}^{-1}$, 300 mm (D) $0.05 \ln 3 \text{ min}^{-1}$, 200 mm

D-6. Which integrated equation is correct for the following 1st order reaction started with only A(g) in a closed rigid vessel.



P_i = initial pressure ; P_t = total pressure at time t

- (A) $K = \frac{2.303}{t} \log_{10} \left[\frac{P_i}{P_t} \right]$ (B) $K = \frac{2.303}{t} \log_{10} \left[\frac{P_t}{P_i} \right]$
(C) $K = \frac{2.303}{t} \log_{10} \left[\frac{2P_i}{3P_i - P_t} \right]$ (D) $K = \frac{2.303}{t} \log_{10} \left[\frac{3P_i}{2P_i - 3P_t} \right]$

D-7. The decomposition NH_3 gas on a heated tungsten surface gave the following results :

Initial pressure (mm of Hg)	65	105	y	185
Half-life (sec)	290	x	670	820

Calculate approximately the values of x and y.

- (A) $x = 410 \text{ sec}$, $y = 115 \text{ mm of Hg}$ (B) $x = 467 \text{ sec}$, $y = 150 \text{ mm of Hg}$
(C) $x = 490 \text{ sec}$, $y = 120 \text{ mm of Hg}$ (D) $x = 430 \text{ sec}$, $y = 105 \text{ mm of Hg}$

D-8. In the reaction $NH_4NO_2(aq.) \rightarrow N_2(g) + 2H_2O(l)$ the volume of N_2 after 20 min and after a long time is 40 ml and 70 ml respectively. The value of rate constant is :

- (A) $(1/20) \log (7/4) \text{ min}^{-1}$ (B) $(2.303/1200) \log (7/3) \text{ sec}^{-1}$
(C) $(1/20) \log (7/3) \text{ min}^{-1}$ (D) $(2.303/20) \log (11/7) \text{ min}^{-1}$

Section (E) : Method to monitor the progress of Reactions (Titration Method and Optical Activity Method)

E-1. Half life of reaction : $H_2O_2(aq) \longrightarrow H_2O(l) + \frac{1}{2} O_2(g)$ is independent of initial concentration of H_2O_2 volume of O_2 gas after 20 minute is 5L at 1 atm and 27°C and after completion of reaction 50L. The rate constant is

- (A) $\frac{1}{20} \log 10 \text{ min}^{-1}$ (B) $\frac{2.303}{20} \log 10 \text{ min}^{-1}$ (C) $\frac{2.303}{20} \log \frac{50}{45} \text{ min}^{-1}$ (D) $\frac{2.303}{20} \log \frac{45}{50} \text{ min}^{-1}$



E-2. The decomposition of N_2O_5 in chloroform was followed by measuring the volume of O_2 gas evolved : $2\text{N}_2\text{O}_5(\text{CCl}_4) \rightarrow 2\text{N}_2\text{O}_4(\text{CCl}_4) + \text{O}_2(\text{g})$. The maximum volume of O_2 gas obtained was 100 cm^3 . In 500 minutes, 90 cm^3 of O_2 were evolved. The first order rate constant (in min^{-1}) for the disappearance of N_2O_5 is :

- (A) $\frac{2.303}{500}$ (B) $\frac{2.303}{500} \log \frac{100}{90}$ (C) $\frac{2.303}{500} \log \frac{90}{100}$ (D) $\frac{100}{10 \times 500}$

E-3. For a reaction $\text{A} \longrightarrow \text{B} + \text{C}$, it was found that at the end of 10 minutes from the start the total optical rotation of the system was 50° and when the reaction is complete, it was 100° . Assuming that only B and C are optically active and dextrorotatory. Calculate the rate constant of this first order reaction.

- (A) 0.693 min^{-1} (B) 0.0693 sec^{-1} (C) 0.0693 min^{-1} (D) 0.00693 sec^{-1}

E-4. The following data were obtained in an experiment on inversion of cane sugar (a first order kinetics)

Time (min)	0	10	After a long time
Total angle of rotation (degree)	+40	+15	-10

The rate constant (in second^{-1}) is $[\ln 2 = 0.693]$

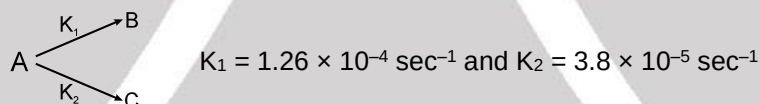
- (A) 0.0693 (B) 1.15×10^{-3} (C) 0.693 (D) 1.15×10^{-2}

E-5. If no catalyst (H^+) is present in acid hydrolysis of ester then rate constant k is : (Where V_0 , V_t and V_∞ are volumes of NaOH used to titrate reaction mixture at $t = 0$, $t = t$ and $t = \infty$)

- (A) $\frac{2.303}{t} \log \frac{V_0}{(V_t - V_\infty)}$ (B) $\frac{2.303}{t} \log \frac{V_\infty}{(V_\infty - V_t)}$
 (C) $\frac{2.303}{t} \log \frac{V_0}{V_t}$ (D) $\frac{2.303}{t} \log \frac{V_\infty}{(V_t - V_\infty)}$

Section (F) : Complication in 1st Order Reaction

F-1. The substance undergoes first order decomposition. The decomposition follows two parallel first order reactions as :



The percentage distribution of B and C

- (A) 80% B and 20% C (B) 76.83% B and 23.17% C
 (C) 90% B and 10% C (D) 60% B and 40% C

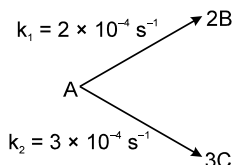
F-2. For a hypothetical elementary reaction,



Initially only 2 moles of A are present. The total number of moles of A, B and C at the end of 75% of reaction are:

- (A) 2 (B) 3 (C) 4 (D) 3.5

F-3. For the following parallel chain reaction



any time is 2M then what will be $[\text{B}]_t$ and $[\text{C}]_t$ respectively ?

- (A) $\frac{11}{12} \text{ M}, \frac{13}{12} \text{ M}$ (B) $\frac{3}{4} \text{ M}, \frac{5}{4} \text{ M}$ (C) $\frac{4}{5} \text{ M}, \frac{6}{5} \text{ M}$ (D) $\frac{8}{13} \text{ M}, \frac{18}{13} \text{ M}$

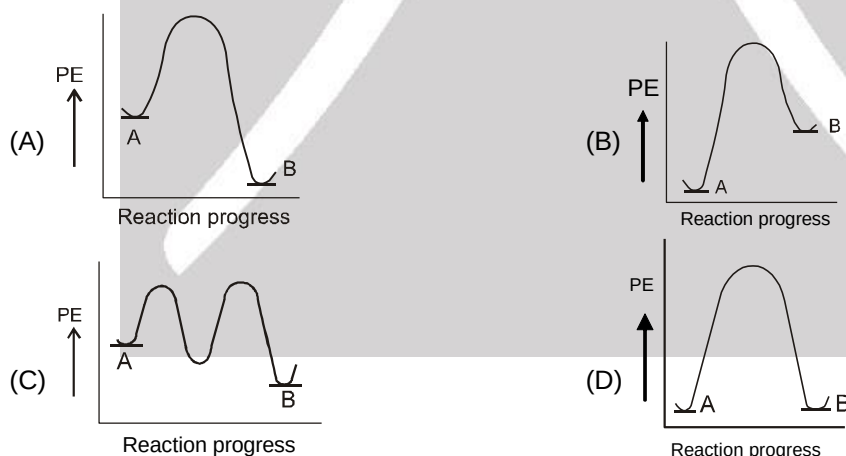


- F-4.** The rate constant for two parallel reactions were found to be $1.0 \times 10^{-2} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and $3.0 \times 10^{-2} \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$. If the corresponding energies of activation of the parallel reactions are 60.0 kJ mol^{-1} and 70.0 kJ mol^{-1} respectively, what is the apparent overall energy of activation ?
 (A) $130.0 \text{ kJ mol}^{-1}$ (B) 67.5 kJ mol^{-1} (C) $100.0 \text{ kJ mol}^{-1}$ (D) 65.0 kJ mol^{-1}

CHEMICAL KINETICS-II

Section (G) : Temperature dependence of rate

- G-1.** The rate constant, the activation energy and the frequency factor of a chemical reaction at 25°C are $3.0 \times 10^{-4} \text{ s}^{-1}$, $104.4 \text{ kJ mol}^{-1}$ and $6.0 \times 10^{14} \text{ s}^{-1}$ respectively. The value of the rate constant as $T \rightarrow \infty$ is :
 (A) $2.0 \times 10^{18} \text{ s}^{-1}$ (B) $6.0 \times 10^{14} \text{ s}^{-1}$ (C) infinite (D) $3.6 \times 10^{30} \text{ s}^{-1}$
- G-2.** The rate constant k_1 of a reaction is found to be double that of rate constant k_2 of another reaction. The relationship between corresponding activation energies of the two reactions at same temperature (E_1 and E_2) will be represented as :
 (A) $E_1 > E_2$ (B) $E_1 < E_2$ (C) $E_1 = E_2$ (D) None of these
- G-3.** When a graph between $\log k$ and $1/T$ is drawn a straight line is obtained. The point at which line cuts y - axis and x -axis respectively correspond to the temp :
 (A) 0, $E_a / 2.303 R \log A$ (B) ∞ , $E_a / (R \ln A)$
 (C) 0, $\log A$ (D) None of these
- G-4.** How much faster would a reaction proceed at 25°C than at 0°C if the activation energy is 65 kJ ?
 (A) 2 times (B) 5 times (C) 11 times (D) 16 times
- G-5.** For a reaction $A \rightarrow B$, $E_a = 10 \text{ kJ mol}^{-1}$, $\Delta H = 5 \text{ kJ mol}^{-1}$. Thus, potential energy profile for this reaction is :

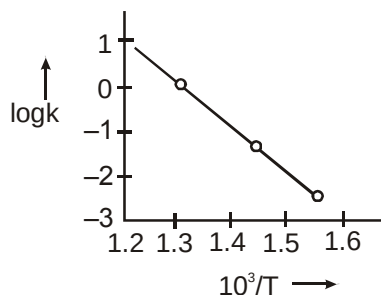


Section (H) : Catalyst dependence of rate and type of Reaction and Determination of rate law and order with the help of given mechanism

- H-1.** The E_a of a reaction in presence of a catalyst is 4.15 kJ mol^{-1} and in absence of a catalyst is 8.3 kJ mol^{-1} .
 1. What is slope of the plot of $\ln k$ vs $\frac{1}{T}$ in the absence of catalyst.
 (A) +1 (B) -1 (C) +1000 (D) -1000



H-2. For the decomposition of HI the following logarithmic plot is shown : [R = 1.98 cal/mol-K]



The activation energy of the reaction is about

- (A) 45600 cal (B) 13500 cal (C) 24600 cal (D) 32300 cal

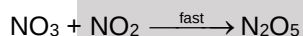
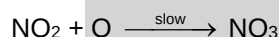
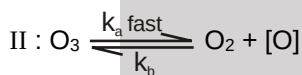
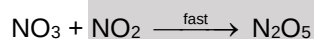
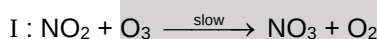
H-3. For the reaction $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightarrow 2\text{HBr}(\text{g})$ the experiment data suggested that $r = k[\text{H}_2][\text{Br}_2]^{1/2}$
The molecularity and order of the reaction are respectively :

- (A) 2, 3/2 (B) 3/2, 3/2 (C) Not defined, 3/2 (D) 1, 1/2

H-4. The reaction of $\text{NO}_2(\text{g})$ and $\text{O}_3(\text{g})$ is first-order in $\text{NO}_2(\text{g})$ and $\text{O}_3(\text{g})$



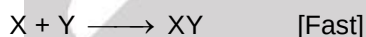
The reaction can take place by mechanism :



Select correct mechanism.

- (A) I only (B) II only (C) both I and II (D) None of I and II

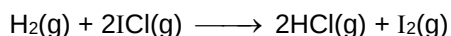
H-5. A hypothetical reaction $\text{X}_2 + \text{Y}_2 \longrightarrow 2\text{XY}$ follows the mechanism given below.



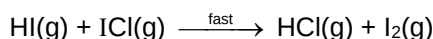
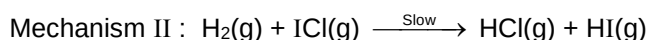
The order of overall reaction is

- (A) 2 (B) 1 (C) 1.5 (D) Zero

H-6. The reaction of hydrogen, and iodine monochloride is represented by the equation :



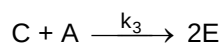
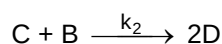
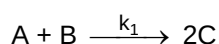
This reaction is first-order in $\text{H}_2(\text{g})$ and also first-order in $\text{ICl}(\text{g})$. Which of these proposed mechanism can be consistent with the given information about this reaction ?



- (A) I only (B) II only (C) both I and II (D) neither I nor II



H-7. Reaction $A + B \longrightarrow D + E$ take place as



The rate of disappearance of C is given by

$$(A) -\frac{d[C]}{dt} = k_2[B][C] + k_3[A][C] - 2k_1[A][B]$$

$$(B) -\frac{d[C]}{dt} = k_2[B][C] + k_3[E] - k_1[C]$$

$$(C) -\frac{d[C]}{dt} = k_2[D] + k_3[E] - k_1[C]$$

$$(D) -\frac{d}{dt} [C] = 2k_1[A][B] - k_2[B][C] - k_3[A][C]$$

RADIO ACTIVITY

Section (I) : Radio Activity

I-1. Radioactivity is affected by :

(A) temperature

(B) Pressure

(C) electric and magnetic field

(D) none of these

I-2. Half life period of 10 gm radioactive element is 20 days. Then half life period of 100 gm of this element is :

(A) 20 days

(B) 200 days

(C) 100 days

(D) 10 days

I-3. Determine the average life of U^{238} having $t_{0.5} = 138.5$ day.

(A) 138.5 day

(B) 199.86 day

(C) 238 day

(D) None of these

I-4. A radioactive element has a half life of one day. After three days the amount of the element left will be :

(A) $1/2$ of the original amount

(B) $1/4$ of the original amount

(C) $1/8$ of the original amount

(D) $1/16$ of the original amount

I-5. A sample of rock from moon contains equal number of atoms of uranium and lead ($t_{1/2}$ for U = 4.5×10^9 years). The age of the rock would be :

(A) 9.0×10^9 years

(B) 4.5×10^9 years

(C) 13.5×10^9 years

(D) 2.25×10^9 years

I-6. The radioactive decay is first order reaction. Its rate, on increasing 10°C temperature

(A) increases 2-3 times

(B) decreases 2-3 times

(C) does not change

(D) increases 10 times

I-7. The half – life of a radioactive isotope is 3 hours . If the initial mass of the isotope were 256 gm , the mass of it remaining undecayed after 18 hours would be :

(A) 4 gm

(B) 8 gm

(C) 12 gm

(D) 16 gm

I-8. A radioactive isotope X with half - life of 6.93×10^9 years decay to Y which is stable. A sample of rock from the moon was found to contain both the element X and Y in the mole ratio 1 : 7. What is the age of the rock?

(A) 2.079×10^{10} years

(B) 1.94×10^{10} years

(C) 1.33×10^9 years

(D) 10^{10} years



- I-9. The half-life of ^{99}Tc is 6.0 h. The total residual activity in a patient after receiving an injection containing ^{99}Tc must not be more than $0.01 \mu\text{Ci}$, after 36.0 hours. What is the maximum activity (in μCi) that the sample injected can have ?
 (A) 0.06 (B) 0.16 (C) 0.32 (D) 0.64
- I-10. The activity per mL of a solution of radioactive substance is x. How much water be added to 200 mL of this solution so that the activity falls to $x/20$ per mL after 4 half-lives ?
 (A) 100 mL (B) 150 mL (C) 80 mL (D) 50 mL

PART - III : MATCH THE COLUMN

1. Match the following :

	Column-I		Column-II
	(Graph)		(Slope)
(A)	c vs t (abscissa) for zero order	(p)	unity
(B)	$\log c$ vs t (abscissa) for first order	(q)	zero
(C)	$\left(\frac{-dc}{dt}\right)$ vs c for zero order	(r)	-k
(D)	$\ln\left(\frac{-dc}{dt}\right)$ vs $\ln c$ for first order	(s)	$-\frac{k}{2.303}$

2. Match the following:

	Column-I		Column-II
(A)	$A + B \longrightarrow C + D$ $r = k_1 [A] [B]$	(p)	Unit of rate constant possess concentration unit
(B)	$A + B \longrightarrow C + D$ $r = k_2 [A] [B]^0$	(q)	Rate constant for the reaction of both the reactants are equal
(C)	$A + B \longrightarrow C + D$ $r = k_3 [A]^0 [B]^0$	(r)	Rate of consumption of at least one of the reactants is equal to rate of production of at least one of the products
(D)	$2A + B \longrightarrow 2C + 3D$ (s) $r = k_3 [A]^0 [B]^0$	(s)	If both reactants are taken in stoichiometric ratio, half life for both reactants are equal.

3. Match the following :

	Column-I		Column-II
(A)	If the activation energy is 65 kJ then how much time faster a reaction proceed at 25°C than at 0°C	(p)	2
(B)	Rate constant of a first-order reaction is 0.0693 min^{-1} . If we start with 20 mol L^{-1} , it is reduced to 2.5 mol L^{-1} in how many minutes.	(q)	Zero
(C)	Half-lives of first-order and zero order reactions are same. Ratio of rates at the start of reaction is how many times of 0.693. Assume initial concentration to be same for the both.	(r)	11
(D)	The half-life periods are given , $[A]_0$ (M) 0.0677 0.136 0.272 $t_{1/2}$ (sec) 240 480 960 order of the reaction is	(s)	30



Exercise-2

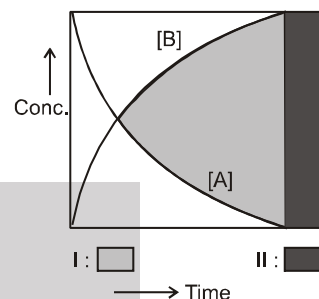
Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

1. In the following graphical representation for the reaction

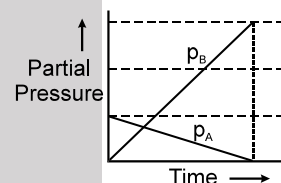
$A \longrightarrow B$ there are two types of regions :

- (A) I and II both represent kinetic region at different interval
 (B) I and II both represent equilibrium regions at different time interval
 (C) I represents kinetic while II represents equilibrium region
 (D) I represents equilibrium while II represents kinetic region



2. If for a reaction in which $A(g)$ converts to $B(g)$ the reaction carried out at const. V & T results into the following graph.

- (A) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a first order reaction.
 (B) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a second order reaction.
 (C) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a zero order reaction.
 (D) then the reaction must be $A(g) \leftrightarrow 3B(g)$ and is a first order reaction.



3. For the first-order reaction T_{av} (average life), T_{50} and T_{75} in the increasing order are :
 (A) $T_{50} < T_{av} < T_{75}$ (B) $T_{50} < T_{75} < T_{av}$ (C) $T_{av} < T_{50} < T_{75}$ (D) $T_{av} = T_{50} < T_{75}$

4. Concentration of the reactant in first-order is reduced to $\frac{1}{e^2}$ of initial concentration after :

(Natural life = $\frac{1}{K}$)

- (A) one natural life-time (B) two-natural life-time
 (C) three natural life-time (D) four natural life-time

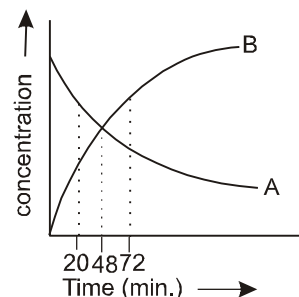
5. For a chemical reaction $A \rightarrow$ products, the rate of disappearance of A is given by :

$\frac{-dC_A}{dt} = \frac{K_1 C_A}{1 + K_2 C_A}$ At low C_A the reaction is of the order with rate constant.....(Assume K_1, K_2 are lesser than 1)

- (A) I, $\frac{K_1}{K_2}$ (B) I, K_1 (C) II, K_1/K_2 (D) II, $\frac{K_1}{K_1 + K_2}$

6. For a first order reaction, $nA \longrightarrow B$ whose concentration vs time curve is as shown in the figure. If half life for the reaction is 24 minutes . Find out the value of n.

- (A) 1
 (B) 2
 (C) 3
 (D) 4





7. In a reaction involving one single reactant, the fraction of the reactant consumed may be defined as $f = \left(1 - \frac{C}{C_0}\right)$ where C_0 and C are the concentrations of the reactant at the start and after time, t . For a first order reaction

(A) $\frac{df}{dt} = k(1 - f)$ (B) $-\frac{df}{dt} = kf$ (C) $-\frac{df}{dt} = k(1 - f)$ (D) $\frac{df}{dt} = kf$

8. $A \rightarrow \text{Product}$ and $\left(\frac{dx}{dt}\right) = k[A]^2$. If $\log\left(\frac{dx}{dt}\right)$ is plotted against $\log[A]$, then graph is of the type :

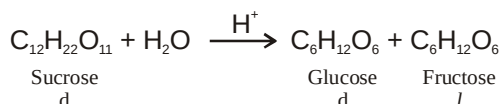


9. The following data is for the decomposition of ammonium nitrite in aqueous solution.

Volume of N_2 produced in cc.	Time (minutes)
7	10
12.6	20
35	infinity

The order of the reaction is

- (A) zero (B) one (C) – one (D) – two
10. Formation of NO_2F from NO_2 and F_2 as per the reaction $2NO_2(g) + F_2(g) \longrightarrow 2NO_2F(g)$ is a second order reaction, first order with respect to NO_2 and first order with respect to F_2 . If NO_2 and F_2 are present in a closed vessel in ratio 2 : 1 maintained at a constant temperature with an initial total pressure of 3 atm, what will be the total pressure in the vessel after the reaction is complete?
- (A) 1 atm (B) 2 atm (C) 2.5 atm (D) 3 atm
11. The inversion of cane sugar proceeds with half life of 50 minute at $pH = 5$ for any concentration of sugar. However if $pH = 6$, the half life changes to 500 minute of any concentration of sugar. The rate law expression for the inversion of cane sugar is written as : $r = k [\text{sugar}]^1 [H^+]^y$. Determine the value of y .
- (A) $r = k [\text{sugar}]^1 [H^+]^{-1}$ (B) $r = k [\text{sugar}]^1 [H^+]^2$ (C) $r = k [\text{sugar}]^1 [H^+]^1$ (D) $r = k [\text{sugar}]^1 [H^+]^0$.
12. Inversion of sucrose ($C_{12}H_{22}O_{11}$) is first-order reaction and is studied by measuring angle of rotation at different instant of time

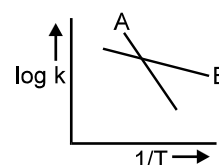
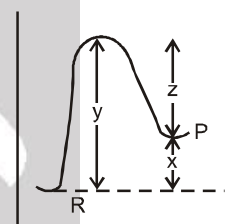
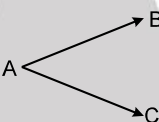
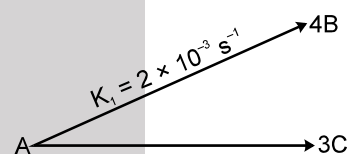


If $(r_\infty - r_0) = a$ and $(r_\infty - r_t) = (a - x)$ (where r_0 , r_t and r_∞ are the angle of rotation at the start, at the time t and at the end of the reaction respectively, then there is 50% inversion when :

(A) $r_0 = 2r_t - r_\infty$ (B) $r_0 = r_t - r_\infty$ (C) $r_0 = r_t - 2r_\infty$ (D) $r_0 = r_t + r_\infty$

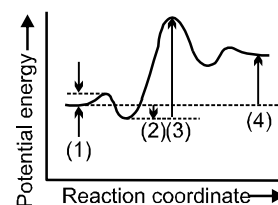
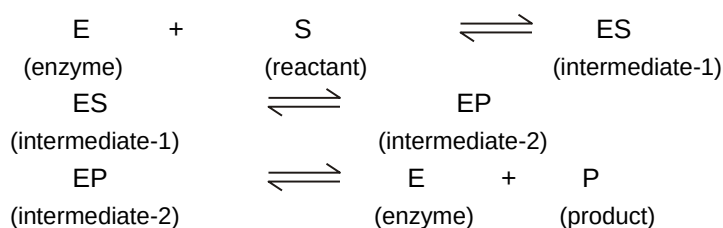


13. In a hypothetical reaction, $A(aq) \longrightarrow 2B(aq) + C(aq)$ (1st order decomposition)
'A' is optically active (dextro-rotatory) while 'B' and 'C' are optically inactive but 'B' takes part in a titration reaction (fast reaction) with H_2O_2 . Hence, the progress of reaction can be monitored by measuring rotation of plane polarised light or by measuring volume of H_2O_2 consumed in titration.
In an experiment the optical rotation was found to be $\theta = 40^\circ$ at $t = 20$ min and $\theta = 10^\circ$ at $t = 50$ min. from start of the reaction. If the progress would have been monitored by titration method, volume of H_2O_2 consumed at $t = 15$ min. (from start) is 40 ml then volume of H_2O_2 consumed at $t = 60$ min will be:
(A) 60 ml (B) 75 ml (C) 52.5 ml (D) 90 ml
14. A.G.M. counter is used to study the radioactive process of first-order. In absence of radioactive substance A, it counts 3 disintegration per second (dps). When A is placed in the G.M. counter, it records 23 dps at the start and 13 dps after 10 minutes. It records x dps after next 10 minutes and A has half-life period y minutes. x and y are :
(A) 8 dps, 10 min (B) 5 dps, 10 min (C) 5 dps, 20 min (D) 5 dps, 5 min
15. For the following parallel chain reaction what will be that value of overall half-life of A in minutes? $\left[\text{Given that } \frac{[B]_t}{[C]_t} = \frac{16}{9} \right]$
(A) 3.3 (B) 6.3 (C) 3.6 (D) None of these
16. For the following parallel chain reaction $A \begin{cases} \rightarrow B \\ \rightarrow C \end{cases}$ the overall half life of A is 12 hours. If rate of formation of C is 60% of a rate of decomposition of A then what will be half life of A while it is converting into B ?
(A) 40 hours (B) 60 hours (C) 50 hours (D) 30 hours
17. The potential energy diagram for a reaction $R \longrightarrow P$ is given below :
 ΔH° of the reaction corresponds to the energy –
(A) x (B) y (C) z (D) (x + y)
18. In gaseous reactions important for the understanding of the upper atmosphere H_2O and O react bimolecularly to form two OH radicals. ΔH for this reaction is 72 kJ at 500 K and E_a is 77 kJ mol⁻¹, then E_a for the bimolecular recombination of two OH radicals to form H_2O and O is :
(A) 3 kJ mol⁻¹ (B) 4 kJ mol⁻¹ (C) 5 kJ mol⁻¹ (D) 7 kJ mol⁻¹
19. The Arrhenius relationship of two different reactions is shown below. Which reaction is faster at a lower temperature and which is more sensitive to changes of temperature ?
(A) B faster, A more sensitive (B) B in both cases
(C) A in both cases (D) A faster, B more sensitive





20. For the same mechanism as Problem refer to the given Fig. and choose the correct set of identifications.



	(1)	(2)	(3)	(4)
(A)	ΔE for $\text{E} + \text{S} \rightarrow \text{ES}$	E_a for $\text{ES} \rightarrow \text{EP}$	$\Delta E_{\text{overall}}$ for $\text{S} \rightarrow \text{P}$	E_a for $\text{EP} \rightarrow \text{E} + \text{P}$
(B)	E_a for $\text{E} + \text{S} \rightarrow \text{ES}$	ΔE for $\text{E} + \text{S} \rightarrow \text{ES}$	E_a for $\text{ES} \rightarrow \text{EP}$	$\Delta E_{\text{overall}}$ for $\text{S} \rightarrow \text{P}$
(C)	E_a for $\text{ES} \rightarrow \text{EP}$	E_a for $\text{EP} \rightarrow \text{E} + \text{P}$	$\Delta E_{\text{overall}}$ for $\text{S} \rightarrow \text{P}$	ΔE for $\text{EP} \rightarrow \text{E} + \text{P}$
(D)	E_a for $\text{E} + \text{S} \rightarrow \text{ES}$	E_a for $\text{ES} \rightarrow \text{EP}$	E_a for $\text{EP} \rightarrow \text{E} + \text{P}$	$\Delta E_{\text{overall}}$ for $\text{S} \rightarrow \text{P}$
(E)	ΔE for $\text{E} + \text{S} \rightarrow \text{ES}$	$\Delta E_{\text{overall}}$ for $\text{S} \rightarrow \text{P}$	ΔE for $\text{EP} \rightarrow \text{E} + \text{P}$	E_a for $\text{EP} \rightarrow \text{E} + \text{P}$

PART - II : NUMERICAL VALUE TYPE

- In a catalyst experiment involving the Haber process $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, the rate of reaction was measured.

$$\text{Rate} = \frac{\Delta[\text{NH}_3]}{\Delta t} = 2.0 \times 10^{-4} \text{ mol L}^{-1} \text{ S}^{-1}.$$
 The rate of reaction of N_2 is $p \times 10^{-4}$. Then find the value of p .
- A substance A is mixed with equal quantities of the substance B and C. At the end of 1000 second, half of the amount of A has reacted. What fraction of A will be left unreacted at the end of 2000 second. If the reaction is zero order with respect to A ?
- For a first order reaction, time required for 99.0% completion is x times for the time required for the completion of 90% of the reaction x is :
- A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives are 54.0 min for A and 18.0 min for B. If the concentrations of A and B are equal initially, how long (in minute) will it take for the concentration of A to be four times that of B ?
- At 80% of a radioactive element undergoing decay is left over after a certain period of time t from pure start, how many such periods should elapse from the start for just over 50% of the element to be left over.
- In a II order reaction, when the concentration of both the reactants are equal, the reaction is completed 20% in 500 sec. How long it would take for the reaction to go to 60% completion ? (Report your answer terms of minutes).
- The acid catalysed hydrolysis of an organic compound A at 30°C has a time for half change of 100 minutes, when carried out in a buffer solution at pH 5 and of 10 minutes, when carried out at pH 4. Both times of half change are independent of the initial concentration of A. If the rate of reaction is given by $-\frac{d[A]}{dt} = k[A]^a[H^+]^b$, what are the values of $(a + b)$?



8. The rate of decomposition of a substance increases by a factor 2.25 for 1.5 times increase in concentration of substance at same temperature. Find out order of reaction.
9. Half-life period for decomposition of NH_3 over tungsten wire are given below :

Initial Pressure in min	50	100	200
$T_{1/2}$	3.52	1.82	0.93

Calculate order of reaction.

10. The rate constant for the II order neutralization of 2-nitropropane by aqueous alkali obeys the equation

$$\log_{10} K = \frac{3163}{T} + 12 \quad T \text{ is temperature in Kelvin.}$$

Time and concentration were in minute and mol litre^{-1} respectively. If half life at 43.3°C and for an initial concentration of $0.001 \text{ mol litre}^{-1}$ for each of the reactant is. Expressed as 10^{-x} , then x is :

11. The ratio of activities of two radio nuclides X and Y in a mixture at time $t = 0$ was found to be 4 : 1. After two hours, the ratio of activities become 1 : 1. If the $t_{1/2}$ of radio nuclide X is 20 min then $t_{1/2}$ [in minutes] of radio nuclide Y is :

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

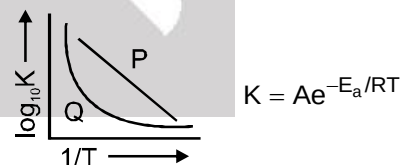
1. Which of the following are true about rate of a reaction?
 (A) Rate of a reaction cannot be negative
 (B) Rate of a reaction of a reactant is change in concentration of the reactant per unit time.
 (C) Average rate and instantaneous rate can never be equal.
 (D) Rate depends upon surface area of the reactants.
2. Which of the following statement is/are correct?
 (A) A second order reaction must be a bimolecular elementary reaction
 (B) A bimolecular elementary reaction must be a second order reaction
 (C) Order of reaction can't be predicted by stoichiometry of reaction.
 (D) First order reaction may be complex or elementary reaction
3. For the reaction $2\text{A} + \text{B} \longrightarrow \text{C}$ with the rate law $\frac{d[\text{C}]}{dt} = k [\text{A}]^1 [\text{B}]^{-1}$ and started with A and B in stoichiometric proportion. Which is/are true?
 (A) unit of k is Ms^{-1}
 (B) [A], [B] and [C] all will be linear functions of time
 (C) $[\text{C}] = 2kt$
 (D) $[\text{C}] = kt$
4. Decomposition of $3\text{A(g)} \longrightarrow 2\text{B(g)} + 2\text{C(g)}$ follows first order kinetics. Initially only A is present in the container. Pressure developed after 20 min. and infinite time are 3.5 and 4 atm respectively. Which of the following is true.
 (A) $t_{50\%} = 20 \text{ min}$ (B) $t_{75\%} = 40 \text{ min}$ (C) $t_{99\%} = 64/3 \text{ min}$ (D) $t_{87.5\%} = 60 \text{ min}$
5. In the following gaseous phase first order reaction, $\text{A(g)} \longrightarrow 2\text{B(g)} + \text{C(g)}$
 initial pressure was found to be 400 mm and it changed to 1000 mm after 20 min. Then :
 (A) Half life for A is 10 min
 (B) Rate constant is 0.0693 min^{-1}
 (C) Partial pressure of C at 30 min is 350 mm
 (D) Total pressure after 30 min is 1100 mm



6. For a second order reaction plots are made for $\frac{1}{[A]}$ vs time for the reaction, $2A \longrightarrow \text{Product}$. Pick up the correct sentences. [k = Rate constant of A]
 (A) the graph will show straight line with slope k
 (B) the graph will show straight line with intercept $[A]_0$
 (C) the graph will show straight line with slope $[A]_0$
 (D) the graph will show straight line with intercept $\frac{1}{[A]_0}$
7. Taking the reaction, $A + 2B \longrightarrow \text{Products}$, to be of second order, which of the following is/are the correct rate law expression(s) ?
 (A) $\frac{dx}{dt} = k[A]^2$ (B) $\frac{dx}{dt} = k[A][B]^2$ (C) $\frac{dx}{dt} = k[A][B]$ (D) $\frac{dx}{dt} = k_1[A] + k_2[B]^2$
8. For the reaction $CH_4 + Br_2 \longrightarrow CH_3Br + HBr$ the experimental data require the following rate equation :

$$\frac{d}{dt} [CH_3Br] = \frac{k_1[CH_4][Br_2]}{1 + k_2[HBr]/[Br_2]}$$

 Which of the following is/are true regarding this ?
 (A) The reaction is a single step reaction
 (B) The reaction is 2nd order in the initial stages $\{[HBr] \approx 0\}$
 (C) The reaction is 2nd order in the final stages $\{[Br_2] \approx 0\}$
 (D) The molecularity of the reaction is two
9. Which of the following statements about the arrhenius equation are correct ?
 (A) The pre exponential factor become equal to the rate constant of the reaction at temperature $T \rightarrow \infty$
 (B) When the activation energy of the reaction $\rightarrow$ zero, the rate becomes independent of temperature.
 (C) The term $e^{-E_a/RT}$ represents the fractions of the molecules having energy is excess of the threshold value.
 (D) On raising temperature , rate constant of the reaction of greater activation energy increases less rapidly than that of the reaction of smaller activation energy.
10. A large increase in the rate of a reaction for a rise in temperature is due to
 (A) increase in the number of collisions (B) the increase in the number of activated molecules
 (C) The shortening of mean free path (D) the lowering of activation energy
11. Which of the following statement are true regarding the $\log K$ vs. $1/T$ plot shown in the given diagram ?
 (A) Plot P shows that the energy of activation is independent of temperature.
 (B) Plot Q describes the behaviour of temperature dependence of energy of activation.
 (C) Arrhenius behaviour is described by P.
 (D) The slope of curve P gives the value $-\frac{E_a}{R}$.
12. Rate constant k varies with temperature by equation , $\log k(\text{min}^{-1}) = 5 - \frac{2000}{T(K)}$. We can conclude :
 (A) pre-exponential factor A is 5 (B) E_a is 2000 k cal
 (C) pre-exponential factor A is 10^5 (D) E_a is 9.212 k cal





13. Activation energy of a chemical reaction can be determined by :
 (A) evaluating rate constant at standard temperature
 (B) evaluating velocities of reaction at two different temperatures
 (C) evaluating rate constant at two different temperatures
 (D) changing concentration of reactants
14. Select the correct statements :
 (A) the molecularity of an elementary reaction indicates how many reactant molecules take part in the step.
 (B) the rate law of an elementary reaction can be predicted by simply seeing the stoichiometry of reaction.
 (C) the slowest elementary step in sequence of the reactions governs the overall rate of formation of product.
 (D) a rate law is often derived from a proposed mechanism by imposing the steady state approximation or assuming that there is a pre-equilibrium.
15. If the rate of reaction, $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{Pt}} 2\text{SO}_3(\text{g})$ is given by :

$$\text{Rate} = K \frac{[\text{SO}_2]}{[\text{SO}_3]^{1/2}}$$
 which statements are correct :
 (A) The overall order of reaction is $-1/2$
 (B) The overall order of reaction is $+1/2$
 (C) The reaction slows down as the product SO_3 is build up
 (D) The rate of reaction does not depend upon concentration of SO_3 formed
16. Rate of radioactive disintegration ($-dN/dt$) is also known as :
 (A) Activity (B) Radioactivity (C) Half-life period (D) Average life period

PART - IV : COMPREHENSION

Read the following passage carefully and answer the questions.

Comprehension # 1



Initially at $t = 0$ gas A was present along with some amount of gas C. At $t = 0$ mole fraction of gas C is $1/3$. After some time $t = t_1$, total pressure is half of the final total pressure at $t = t_x$ (a very long time). Assume this decomposition is a first order, at a constant temperature. It is also given at $t = t_x$, final total pressure is 35 bar.

1. At $t = t_1$ pressure of gas B is :
 (A) 2.5 bar (B) 1.25 bar (C) 5.0 bar (D) data is insufficient
2. Rate constant (k) = $(\log 64 - \log 49) \text{ s}^{-1}$. Value of t_1 in seconds is :
 (A) 2.15 s (B) 1.5 s (C) 2.3 s (D) 1.15 s
3. Ratio of rate constant at $t = 0$ to $t = t_1$ to $t = t_x$ is :
 (A) 2 : 3 : 4 (B) 1 : 1 : 1 (C) 1 : 3 : 5 (D) 1 : 3 : 5



Comprehension # 2

Set-I (Without catalyst)			
Reaction	Temperature	E (activation)	k
A → B	T ₁ K	Ea ₁	k ₁
A → B	T ₂ K	Ea ₂	k ₂

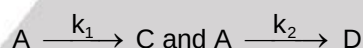
Set-II (With catalyst) (Consider catalyst being positive only)			
Reaction	Temperature	E (activation)	k
A → B	T ₁ K	Ea ₃	k ₃
A → B	T ₂ K	Ea ₄	k ₄

4. For the (Set-1) :
- (A) if T₁ > T₂, k₁ > k₂ always
 (B) if T₁ > T₂, k₁ > k₂ (for exothermic reaction)
 (C) if T₁ > T₂, k₁ < k₂ (for endothermic reaction)
 (D) Ea₁ ≠ Ea₂
5. For the (Set-1) :
- (A) Ea₁ > Ea₂ if T₁ > T₂
 (B) Ea₁ < Ea₂ if T₁ > T₂
 (C) Ea₁ = Ea₂
 (D) Ea₁ = 0.5 Ea₂
6. Comparing set-I and II :
- (A) k₄ > k₃ & k₂ > k₁, if T₂ > T₁ (endothermic)
 (B) k₄ < k₃ & k₂ > k₁, if T₂ < T₁ (endothermic)
 (C) k₄ > k₃ & k₂ > k₁, if T₂ < T₁ (exothermic)
 (D) k₄ < k₃ & k₂ < k₁, if T₂ > T₁ (exothermic)

Comprehension # 3

Competing first-order reactions :

Frequently a species can react in different ways to give a variety of products. For example, toluene can be nitrated at the ortho, meta, or para positions, We shall consider the simplest case, that of two competing irreversible first-order reactions :



where the stoichiometric coefficients are taken as unity for simplicity. The rate law is

$$\left(\frac{d[A]}{dt} \right) = -k_1[A] - k_2[A] = -(k_1 + k_2)[A] \Rightarrow [A] = [A]_0 e^{-(k_1+k_2)t}$$

For C, we have $\left(\frac{d[C]}{dt} \right) = k_1[A] = k_1[A]_0 e^{-(k_1+k_2)t}$. Multiplication by dt and integration from time 0

(where [C]₀ = 0) to an arbitrary time t gives $[C] = \frac{k_1[A]_0}{k_1 + k_2} (1 - e^{-(k_1+k_2)t})$

Similarly, integration of $\left(\frac{d[D]}{dt} \right)$ gives $[D] = \frac{k_2[A]_0}{k_1 + k_2} (1 - e^{-(k_1+k_2)t})$

The sum of the rate constants k₁ + k₂ appears in the exponentials for both [C] and [D].

At any time we also have $\frac{[C]}{[D]} = \frac{k_1}{k_2}$



7. $A \xrightarrow{k_1} 2B$
 $A \xrightarrow{k_2} 3C$
 starting initially with only A Which of the following is correct at time t
 (A) $[A]_0 = [A]_t + [B]_t + [C]_t$ (B) $[A]_0 = [A]_t + 2[B]_t + 3[C]_t$
 (C) $[A]_0 = [A]_t + \frac{[B]_t}{2} + \frac{[C]_t}{3}$ (D) $[A]_0 = \frac{2}{3} [A]_t + [B]_t + [C]_t$
8. $X \xrightarrow{k_1} Y$
 $X \xrightarrow{k_2} Z$
 starting with only 'X', ratio $\frac{[X]_t}{[Y]_t + [Z]_t}$
 (A) Independent of time (B) $\frac{1}{(e^{kt} - 1)}$
 (C) Depends upon initial concentration of X (D) $[A]_0 (e^{kt} - 1)$
9. At high temperature acetic acid decomposes into CO_2 & CH_4 and simultaneously into CH_2CO (ketene) and H_2O
 (i) $CH_3COOH \xrightarrow{k_1=3s^{-1}} CH_4 + CO_2$
 (ii) $CH_3COOH \xrightarrow{k_2=4s^{-1}} CH_2CO + H_2O$
 What is the fraction of acetic acid reacting as per reaction (i) ?
 (A) $\frac{3}{4}$ (B) $\frac{3}{7}$ (C) $\frac{4}{7}$ (D) none of these
10. For $A \xrightarrow{k_1} 3B$
 $A \xrightarrow{k_2} 4C$
 starting with pure A ratio of rate of production of B to C is
 (A) Independent of time (B) Independent of temperature
 (C) Depends upon initial concentration of A (D) Independent of mechanism of reaction

Comprehension # 4

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table.

Columns 1, 2 and 3 contain the Order of the reaction, value of K and half-life respectively.		
Column-1	Column-2	Column-3
(I) First order reaction	(i) $k = \frac{1}{2t} \left[\frac{1}{(a-x)^2} - \frac{1}{a^2} \right]$	(P) Half life $\propto \frac{1}{a^2}$
(II) Second order reaction	(ii) $k = \frac{1}{t} \left[\frac{1}{a-x} - \frac{1}{a} \right]$	(Q) Half life $\propto \frac{1}{a}$
(III) Third order reaction	(iii) $k = \frac{x}{t}$	(R) Half life is doubled on doubling the initial concentration
(IV) Zero order reaction	(iv) $k = \frac{1}{t} \log_e \left(\frac{a}{a-x} \right)$	(S) 50% reaction takes same time even if concentration is halved or doubled.

11. The correct combination for reaction $A \xrightarrow{2 \times 10^{-3} \text{ min}^{-1}} B$ is :
 (A) (II) (ii) (Q) (B) (IV) (iv) (S) (C) (I) (iv) (S) (D) (I) (iii) (R)
12. The incorrect combination among the following is :
 (A) (I) (iii) (S) (B) (II) (ii) (Q) (C) (III) (i) (P) (D) (IV) (iii) (R)
13. For an elementary reaction $2A \longrightarrow B$, correct combination is :
 (A) (I) (iv) (Q) (B) (I) (iv) (S) (C) (II) (ii) (Q) (D) (II) (ii) (R)



Exercise-3

* Marked Questions may have more than one correct option.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

- The rate constant for the reaction, $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$, is $3 \times 10^{-5} \text{ s}^{-1}$. If the rate is $2.4 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$, then the concentration of N_2O_5 (in mol L^{-1}) is : [JEE-2000(S), 1/35]

(A) 1.4 (B) 1.2 (C) 0.04 (D) 0.8
- $^{238}_{92}\text{U}$ is radioactive and it emits α and β particles to form $^{206}_{82}\text{Pb}$. Calculate the number of α and β particles emitted in this conversion. An ore of $^{238}_{92}\text{U}$ is found to contain $^{238}_{92}\text{U}$ and $^{206}_{82}\text{Pb}$ in the weight ratio of 1 : 0.9 the half life period of $^{238}_{92}\text{U}$ is 4.5×10^9 years. Calculate the age of the ore. [JEE-2000(M)]
- A hydrogenation reaction is carried out at 500 K. If the same reaction is carried out in the presence of a catalyst at the same rate, the temperature required is 400 K. Calculate the activation energy of the reaction if the catalyst lowers the activation barrier by 20 kJ mol^{-1} . [JEE-2000(M), 3/100]
- If 'I' is the intensity of absorbed light and 'C' is the concentration of AB for the photochemical process, $\text{AB} + h\nu \rightarrow \text{AB}^*$, the rate of formation of AB^* is directly proportional to [JEE-2001(S), 1/35]

(A) C (B) I (C) I^2 (D) C.I
- The rate of a first order reaction is $0.04 \text{ mole litre}^{-1} \text{ s}^{-1}$ at 10 minutes and $0.03 \text{ mole litre}^{-1} \text{ s}^{-1}$ at 20 minutes after initiation. Find the half life of the reaction. [JEE-2001(M), 5/100]
- Consider the chemical reaction,

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \longrightarrow 2\text{NH}_3(\text{g})$$

The rate of this reaction can be expressed in terms of time derivatives of conc. of $\text{N}_2(\text{g})$, $\text{H}_2(\text{g})$ or $\text{NH}_3(\text{g})$. Identify the correct relationship amongst rate expressions : [JEE-2002(S), 3/90]

(A) $\text{Rate} = -\frac{d[\text{N}_2]}{dt} = -\frac{1}{3}\frac{d[\text{H}_2]}{dt} = \frac{1}{2}\frac{d[\text{NH}_3]}{dt}$ (B) $\text{Rate} = \frac{d[\text{N}_2]}{dt} = -3\frac{d[\text{H}_2]}{dt} = 2\frac{d[\text{NH}_3]}{dt}$

(C) $\text{Rate} = \frac{d[\text{N}_2]}{dt} = -\frac{1}{3}\frac{d[\text{H}_2]}{dt} = \frac{1}{2}\frac{d[\text{NH}_3]}{dt}$ (D) $\text{Rate} = -\frac{d[\text{N}_2]}{dt} = -\frac{d[\text{H}_2]}{dt} = \frac{d[\text{NH}_3]}{dt}$
- ^{64}Cu (half life = 12.8 hours) decays by β^- emission(38%), β^+ emission(19%) and electron capture(43%). Write the decay products and calculate partial half lives for each of the decay processes. [JEE-2002(M), 5/60]
- In the biologically-catalysed oxidation of ethanol, the concentration of ethanol decreases in a first order reaction from 800 mol dm^{-3} to 50 mol dm^{-3} in $2 \times 10^4 \text{ s}$. The rate constant (s^{-1}) of the reaction is : [JEE-2003(S), 3/84]

(A) 3.45×10^{-5} (B) 1.38×10^{-4} (C) 1.00×10^{-4} (D) 5.00×10^{-5}
- Given X $\longrightarrow$ product (Taking 1st order reaction)

conc	0.01	0.0025
(mol/lit)		
time (min)	0	40

Initial rate of reaction is in $\text{mol l}^{-1}\text{min}^{-1}$. [JEE-2004(S), 3/84]

(A) 3.43×10^{-4} (B) 1.73×10^{-4} (C) 3.43×10^{-5} (D) 1.73×10^{-5}



10. Initial rates, r_0 , of the $A + B \rightarrow P$ at different initial concentrations of A and B ($[A]_0$ and $[B]_0$) are given below:

$[A]_0$ (mol L ⁻¹)	$[B]_0$ (mol L ⁻¹)	r_0 (mol L ⁻¹ s ⁻¹)
0.1	0.1	0.05
0.2	0.1	0.10
0.1	0.2	0.05

- (a) Write the rate equation.
 (b) Calculate the rate constant of the reaction.

[JEE-2004(M), 2/60]

11. For a reaction $2X(g) \longrightarrow 3Y(g) + 2Z(g)$ the following data is obtained.

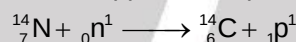
Time (min)	P_x (mm of Hg) (Partial pressure of X)
0	800
100	400
200	200

Find order with respect to X, rate constant of the reaction, time taken for 75% completion and find the total pressure when partial pressure of X, $P_x = 700$ mm of Hg.

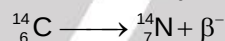
[JEE-2005(M), 2/60]

Comprehension

Carbon-14 is used to determine the age of organic material. The procedure is based on the formation of ^{14}C by neutron capture in the upper atmosphere.



^{14}C is absorbed by living organisms during photosynthesis. The ^{14}C content is constant in living organism, once the plant or animal dies, the uptake of carbon dioxide by it ceases and the level of ^{14}C in the dead being, falls due to the decay which C^{14} undergoes



The half life period of ^{14}C is 5770 years. The decay constant (λ) can be calculated by using the following formula $\lambda = \frac{0.693}{t_{1/2}}$

The comparison of the β^- activity of the dead matter with that of the carbon still in circulation enables measurement of the period of the isolation of the material from the living cycle. The method however, ceases to be accurate over periods longer than 30,000 years. The proportion of ^{14}C to ^{12}C in living matter is 1 : 10^{12} .

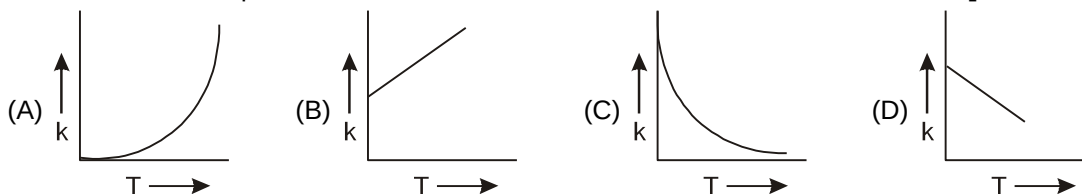
12. Which of the following option is correct? [JEE-2006, 5/184]
 (A) Rate of exchange of carbon between atmosphere and living is slower than decay of ^{14}C .
 (B) Carbon dating can be used to find out the age of earth crust and rocks
 (C) Rate of exchange of ^{14}C between atmosphere and living organism is so fast that an equilibrium is set up between the intake of ^{14}C by organism and its exponential decay.
 (D) Carbon dating can not be used to determine concentration of ^{14}C in dead beings.
13. What should be the age of fossil for meaningful determination of its age? [JEE-2006, 5/184]
 (A) 6 years
 (B) 6000 years
 (C) 60,000 years
 (D) It can be used to calculate any age



14. A nuclear explosion has taken place leading to increase in concentration of ^{14}C in nearby areas. ^{14}C concentration is C_1 in nearby areas and C_2 in areas far away. If the age of the fossil is determined to be T_1 and T_2 at the places respectively then [JEE-2006, 5/184]
- (A) The age of the fossil will increase at the place where explosion has taken place and

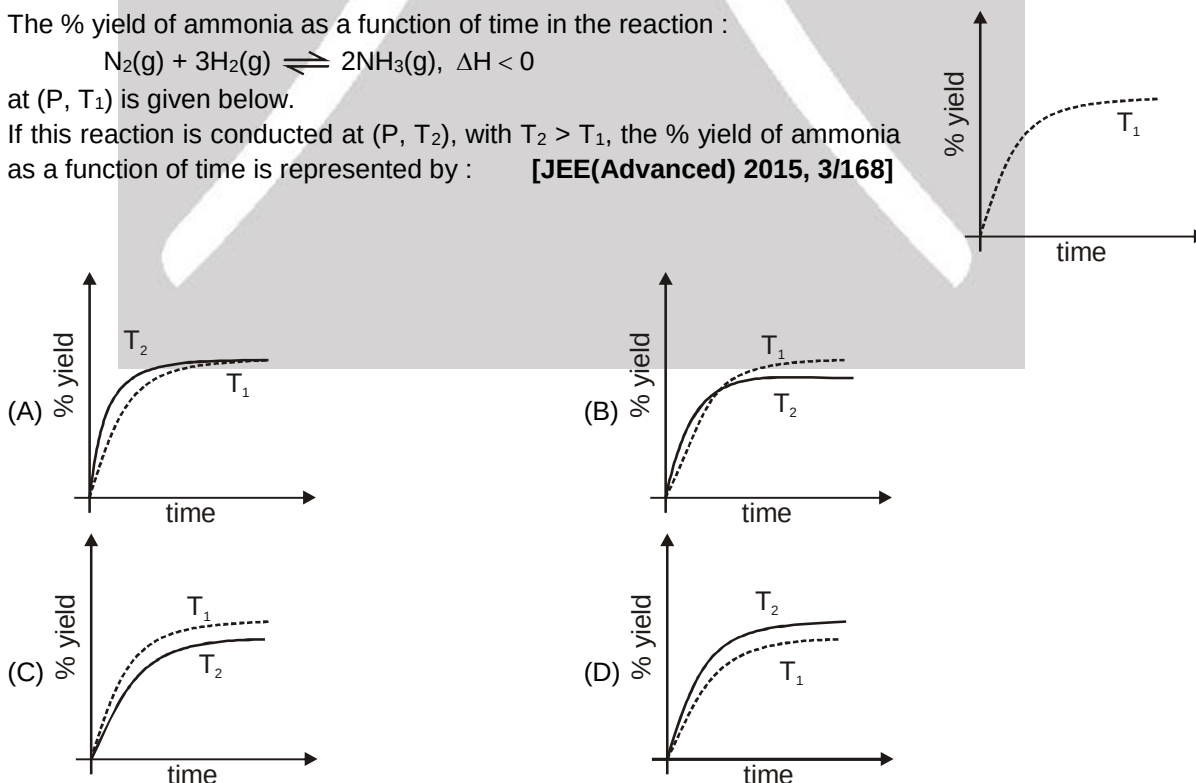
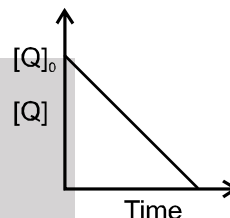
$$T_1 - T_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$
- (B) The age of the fossil will decrease at the place where explosion has taken place and

$$T_1 - T_2 = \frac{1}{\lambda} \ln \frac{C_1}{C_2}$$
- (C) The age of fossil will be determined to be same
- (D) $\frac{T_1}{T_2} = \frac{C_1}{C_2}$
15. Consider a reaction $a\text{G} + b\text{H} \rightarrow \text{Products}$. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is : [JEE-2007, 3/162]
- (A) 0 (B) 1 (C) 2 (D) 3
16. Under the same reaction conditions, initial concentration of $1.386 \text{ mol dm}^{-3}$ of a substance becomes half in 40 seconds and 20 seconds through first order and zero order kinetics, respectively. Ratio $\left(\frac{k_1}{k_0}\right)$ of the rate constant for first order (k_1) and zero order (k_0) of the reaction is. [JEE-2008, 3/162]
- (A) $0.5 \text{ mol}^{-1} \text{ dm}^3$ (B) 1.0 mol dm^{-3} (C) 1.5 mol dm^{-3} (D) $2.0 \text{ mol}^{-1} \text{ dm}^3$
17. For a first order reaction $\text{A} \rightarrow \text{P}$, the temperature (T) dependent rate constant (k) was found to follow the equation $\log k = - (2000) \frac{1}{T} + 6.0$. The pre-exponential factor A and the activation energy E_a , respectively, are : [JEE-2009, 3/160]
- (A) $1.0 \times 10^6 \text{ s}^{-1}$ and 9.2 kJ mol^{-1} (B) 6.0 s^{-1} and 16.6 kJ mol^{-1}
 (C) $1.0 \times 10^6 \text{ s}^{-1}$ and 16.6 kJ mol^{-1} (D) $1.0 \times 10^6 \text{ s}^{-1}$ and 38.3 kJ mol^{-1}
18. The concentration of R in the reaction $\text{R} \rightarrow \text{P}$ was measured as a function of time and the following data is obtained : [JEE-2010, 3/163]
- | | | | | |
|-------------|-----|------|------|------|
| [R] (molar) | 1.0 | 0.75 | 0.40 | 0.10 |
| t(min.) | 0.0 | 0.05 | 0.12 | 0.18 |
- The order of the reaction is :
19. Plots showing the variation of the rate constant (k) with temperature (T) are given below. The plot that follows Arrhenius equation is : [JEE-2010, 3/163]



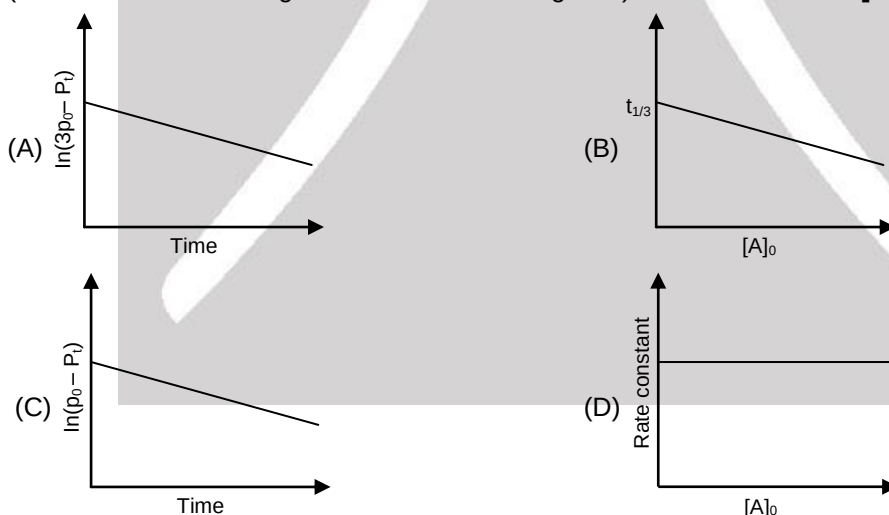


- 20.* For the first order reaction, $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ [JEE-2011, 4/160]
 (A) The concentration of the reactant decreases exponentially with time
 (B) The half-life of the reaction decreases with increasing temperature
 (C) The half-life of the reaction depends on the initial concentration of the reactant
 (D) The reaction proceeds to 99.6% completion in eight half-life duration
21. An organic compound undergoes first-order decomposition. The time taken for its decomposition to $1/8$ and $1/10$ of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{[t_{1/8}]}{[t_{1/10}]} \times 10$?
 ($\log_{10} 2 = 0.3$) [JEE-2012, 4/136]
22. In the reaction, $\text{P} + \text{Q} \longrightarrow \text{R} + \text{S}$
 the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is : [JEE(Advanced) 2013, 2/120]
 (A) 2 (B) 3
 (C) 0 (D) 1
23. The initial rate of hydrolysis of methyl acetate (1M) by a weak acid (HA, 1M) is $1/100^{\text{th}}$ of that of a strong acid (HX, 1M), at 25°C . The K_a of HA is : [JEE(Advanced)-2013, 4/120]
 (A) 1×10^{-4} (B) 1×10^{-5} (C) 1×10^{-6} (D) 1×10^{-3}
24. For the elementary reaction $\text{M} \rightarrow \text{N}$, the rate of disappearance of M increases by a factor of 8 upon doubling the concentration of M. The order of the reaction with respect to M is : [JEE(Advanced) 2014, 3/120]
 (A) 4 (B) 3 (C) 2 (D) 1
25. The % yield of ammonia as a function of time in the reaction :
 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}), \Delta H < 0$
 at (P, T_1) is given below.
 If this reaction is conducted at (P, T_2) , with $\text{T}_2 > \text{T}_1$, the % yield of ammonia as a function of time is represented by : [JEE(Advanced) 2015, 3/168]





26. In dilute aqueous H_2SO_4 , the complex diaquodioxalatoferrate(II) is oxidized by MnO_4^- . For this reaction, the ratio of the rate of change of $[\text{H}^+]$ to the rate of change of $[\text{MnO}_4^-]$ is : [JEE(Advanced)-2015, 4/168]
- 27.* According to the Arrhenius equation, [JEE(Advanced)-2016, 4/124]
 (A) a high activation energy usually implies a fast reaction.
 (B) rate constant increases with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy.
 (C) higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant.
 (D) the pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.
28. In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. the correct option(s) among the following is(are) [JEE(Advanced)-2017, 4/122]
 (A) The activation energy of the reaction is unaffected by the value of the steric factor.
 (B) Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation.
 (C) The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally.
 (D) Since $P = 4.5$, the reaction will not proceed unless an effective catalyst is used.
- 29.* For a first order reaction $\text{A(g)} \rightarrow 2\text{B(g)} + \text{C(g)}$ at constant volume and 300 K, the total pressure at the beginning ($t = 0$) and at time t are P_0 and P_t , respectively. Initially, only A is present with concentration $[\text{A}]_0$, and $t_{1/3}$ is the time required for the partial pressure of A to reach $1/3^{\text{rd}}$ of its initial value. The correct option(s) is (are) [JEE(Advanced)-2018, 4/120]
 (Assume that all these gases behave as ideal gases)



30. Consider the following reversible reaction,

$$\text{A(g)} + \text{B(g)} \rightleftharpoons \text{AB(g)}$$
 The activation energy of the backward reaction exceeds that of the forward reaction by 2 RT (in J mol^{-1}). If the pre-exponential factor of the forward reaction is 4 times that of the reverse reaction, the absolute value of $\Delta G^\ominus$ (in J mole^{-1}) for the reaction at 300 K is _____.
 (Given : $\ln(2) = 0.7$, $\text{RT} = 2500 \text{ J mol}^{-1}$ at 300 K and G is the Gibbs energy)

[JEE(Advanced)-2018, 3/120]



PART - II : JEE (MAIN) ONLINE PROBLEMS (PREVIOUS YEARS)

1. The half-life period of a first order reaction is 15 minutes. The amount of substance left after one hour will be :
 [JEE(Main) 2014 Online (09-04-14), 4/120]
 (1) 1/4 of the original amount (2) 1/8 of the original amount
 (3) 1/16 of the original amount (4) 1/32 of the original amount

2. In the reaction of formation of sulphur trioxide by contact process $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ the rate of reaction was measured as $\frac{d[\text{O}_2]}{dt} = -2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. The rate of reaction in terms of $[\text{SO}_2]$ in $\text{mol L}^{-1} \text{ s}^{-1}$ will be :
 [JEE(Main) 2014 Online (11-04-14), 4/120]
 (1) -1.25×10^{-4} (2) -2.50×10^{-4} (3) -3.75×10^{-4} (4) -5.00×10^{-4}

3. For the reaction, $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$, the rate equation can be expressed in two ways
 $-\frac{d[\text{N}_2\text{O}_5]}{dt} = k[\text{N}_2\text{O}_5]$ and $+\frac{d[\text{NO}_2]}{dt} = k'[\text{N}_2\text{O}_5]$ k and k' related as P:
 [JEE(Main) 2014 Online (11-04-14), 4/120]
 (1) $k = k'$ (2) $2k = k'$ (3) $k = 2k'$ (4) $k = 4k'$

4. The rate coefficient (k) for a particular reactions is $1.3 \times 10^{-4} \text{ M}^{-1}\text{s}^{-1}$ at 100°C , and $1.3 \times 10^{-3} \text{ M}^{-1}\text{s}^{-1}$ at 150°C . What is the energy of activation (E_a) (in kJ) for this reaction ?
 (R = molar gas constant = $8.314 \text{ JK}^{-1} \text{ mol}^{-1}$)
 [JEE(Main) 2014 Online (12-04-14), 4/120]
 (1) 16 (2) 60 (3) 99 (4) 132

5. For the reaction, $3\text{A} + 2\text{B} \rightarrow \text{C} + \text{D}$, the differential rate law can be written as :
 [JEE(Main) 2014 Online (19-04-14), 4/120]
 (1) $\frac{1}{3} \frac{d[\text{A}]}{dt} = \frac{d[\text{C}]}{dt} = k[\text{A}]^n[\text{B}]^m$ (2) $-\frac{d[\text{A}]}{dt} = \frac{d[\text{C}]}{dt} = k[\text{A}]^n[\text{B}]^m$
 (3) $+\frac{1}{3} \frac{d[\text{A}]}{dt} = \frac{d[\text{C}]}{dt} = k[\text{A}]^n[\text{B}]^m$ (4) $-\frac{1}{3} \frac{d[\text{A}]}{dt} = \frac{d[\text{C}]}{dt} = k[\text{A}]^n[\text{B}]^m$

6. The reaction $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ follows first order kinetics. The pressure of a vessel containing only N_2O_5 was found to increase from 50 mmHg to 87.5 mm Hg in 30 min. The pressure exerted by the gases after 60 min. will be (Assume temperature remains constant) :
 [JEE(Main) 2015 Online (10-04-15), 4/120]
 (1) 125 mm Hg (2) 106.25 mm Hg (3) 116.25 mm Hg (4) 150 mm Hg

7. For the equilibrium, $\text{A}(\text{g}) \rightleftharpoons \text{B}(\text{g})$, ΔH is -40 kJ/mol . If the ratio of the activation energies of the forward (E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :
 [JEE(Main) 2015 Online (11-04-15), 4/120]
 (1) $E_f = 80 \text{ kJ/mol}$; $E_b = 120 \text{ kJ/mol}$ (2) $E_f = 60 \text{ kJ/mol}$; $E_b = 100 \text{ kJ/mol}$
 (3) $E_f = 30 \text{ kJ/mol}$; $E_b = 70 \text{ kJ/mol}$ (4) $E_f = 70 \text{ kJ/mol}$; $E_b = 30 \text{ kJ/mol}$

8. $\text{A} + 2\text{B} \rightarrow \text{C}$, the rate equation for this reaction is given as
 Rate = $k[\text{A}][\text{B}]$.
 If the concentration of A is kept the same but that of B is doubled what will happen to the rate itself ?
 [JEE(Main) 2015 Online (11-04-15), 4/120]
 (1) halved (2) the same (3) doubled (4) quadrupled



9. The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process show below:

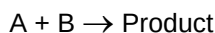


The closest rate constant for the overall reaction $\text{O}_3(\text{g}) + \text{O}^\bullet(\text{g}) \rightarrow 2\text{O}_2(\text{g})$ is:

[JEE(Main) 2016 Online (09-04-16), 4/120]

- (1) $1.4 \times 10^{20} \text{ L mol}^{-1} \text{ s}^{-1}$ (2) $5.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$
 (3) $3.1 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$ (4) $2.6 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$

10. The rate law for the reaction below is given by the expression $k [\text{A}][\text{B}]$



If the concentration of B is increased from 0.1 to 0.3 mole, keeping the value of A at 0.1 mole, the rate constant will be :

[JEE(Main) 2016 Online (10-04-16), 4/120]

- (1) 9 k (2) 3 k (3) k/3 (4) k

11. The rate of a reaction A doubles on increasing the temperature from 300 to 310 K. By how much, the temperature of reaction B should be increased from 300 K so that rate doubles if activation energy of the reaction B is twice to that of reaction A.

[JEE(Main) 2017 Online (08-04-17), 4/120]

- (1) 9.84 K (2) 19.67 K (3) 2.45 K (4) 4.92 K

12. The rate of a reaction quadruples when the temperature changes from 300 to 310 K. The activation energy of this reaction is :

(Assume activation energy and pre-exponential factor are independent of temperature; $\ln 2 = 0.693$; $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

[JEE(Main) 2017 Online (09-04-17), 4/120]

- (1) 53.6 kJ mol⁻¹ (2) 26.8 kJ mol⁻¹ (3) 107.2 kJ mol⁻¹ (4) 214.4 kJ mol⁻¹

13. N_2O_5 decomposes to NO_2 and O_2 and follows first order kinetics. After 50 minutes, the pressure inside the vessel increases from 50 mm Hg to 87.5 mm Hg. The pressure of the gaseous mixture after 100 minute at constant temperature will be :

[JEE(Main) 2018 Online (15-04-18), 4/120]

- (1) 136.25 mm Hg (2) 106.25 mm Hg (3) 175.0 mm Hg (4) 116.25 mm Hg

14. For a first order reaction, $\text{A} \rightarrow \text{P}$, $t_{1/2}$ (half-life) is 10 days. The time required for $\frac{1}{4}$ th conversion of A (in days) is : ($\ln 2 = 0.693$, $\ln 3 = 1.1$)

[JEE(Main) 2018 Online (15-04-18), 4/120]

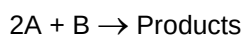
- (1) 5 (2) 3.2 (3) 4.1 (4) 2.5

15. If 50 % of a reaction occurs in 100 second and 75 % of the reaction occurs in 200 second, the order of this reaction is :

[JEE(Main) 2018 Online (16-04-18), 4/120]

- (1) 2 (2) 3 (3) Zero (4) 1

16. The following results were obtained during kinetic studies of the reaction;



Experiment	[A] (in mol L ⁻¹)	[B] (in mol L ⁻¹)	Initial Rate of reaction (in mol L ⁻¹ min ⁻¹)
I	0.10	0.20	6.93×10^{-3}
II	0.10	0.25	6.93×10^{-3}
III	0.20	0.30	1.386×10^{-2}

Time time (in minutes) required to consume half of A is : [JEE(Main) 2019 Online (09-01-19), 4/120]

- (1) 1 (2) 5 (3) 10 (4) 100



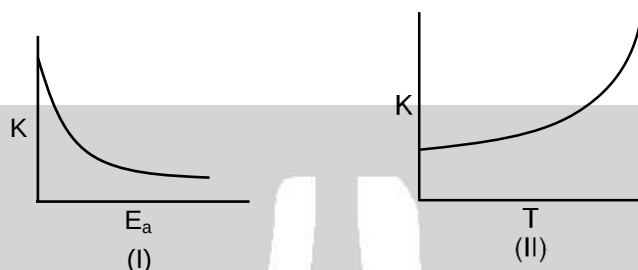
17. For the reaction, $2A + B \rightarrow \text{products}$, when the concentration of A and B both were doubled, the rate of the reaction increased from $0.3 \text{ mol L}^{-1}\text{s}^{-1}$ to $2.4 \text{ mol L}^{-1}\text{s}^{-1}$. When the concentration of A alone is doubled, the rate increased from $0.3 \text{ mol L}^{-1}\text{s}^{-1}$ to $0.6 \text{ mol L}^{-1}\text{s}^{-1}$.

Which one of the following statements is correct?

[JEE(Main) 2019 Online (09-01-19), 4/120]

- (1) Order of the reaction with respect to B is 1 (2) Order of the reaction with respect to B is 2
(3) Total order of the reaction is 4 (4) Order of the reaction with respect to A is 2

18. Consider the given plots for a reaction obeying Arrhenius equation ($0^\circ\text{C} < T < 300^\circ\text{C}$) : (k and E_a are rate constant and activation energy, respectively)



Choose the correct option :

[JEE(Main) 2019 Online (10-01-19), 4/120]

- (1) I is right but II is wrong (2) Both I and II are wrong
(3) I is wrong but II is right (4) Both I and II are correct

19. For an elementary chemical reaction, $A_2 \xrightleftharpoons[k_{-1}]{k_1} 2A$ the expression for $\frac{d[A]}{dt}$

[JEE(Main) 2019 Online (10-01-19), 4/120]

- (1) $2k_1[A_2] - k_{-1}[A]^2$ (2) $k_1[A_2] + k_{-1}[A]^2$ (3) $k_1[A_2] - k_{-1}[A]^2$ (4) $2k_1[A_2] - 2k_{-1}[A]^2$

20. If a reaction follows the Arrhenius equation the plot $\ln k$ vs $1/(RT)$ gives straight line with a gradient ($-y$) unit. The energy required to activate the reactant is :

[JEE(Main) 2019 Online (11-01-19), 4/120]

- (1) $-y$ unit (2) y unit (3) y/R unit (4) yR unit

21. The reaction $2X \rightarrow B$ is a zeroth order reaction. If the initial concentration of X is 0.2 M, the half-life is 6 h. When the initial concentration of X is 0.5 M, the time required to reach its final concentration of 0.2 M will be :

[JEE(Main) 2019 Online (11-01-19), 4/120]

- (1) 18.0 h (2) 12.0 h (3) 7.2 h (4) 9.0 h

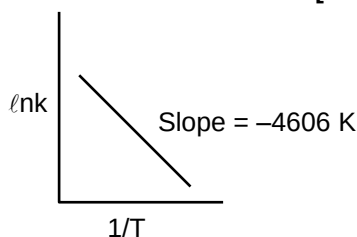
22. Decomposition of X exhibits a rate constant of $0.05 \text{ } \mu\text{g/year}$. How many years are required for the decomposition of $5 \text{ } \mu\text{g}$ of X into $2.5 \text{ } \mu\text{g}$?

[JEE(Main) 2019 Online (12-01-19), 4/120]

- (1) 40 (2) 25 (3) 20 (4) 50

23. For a reaction, consider the plot of $\ln k$ versus $1/T$ given in the figure. If the rate constant of this reaction at 400 K is 10^{-5} s^{-1} , then the rate constant at 500 K is:

[JEE(Main) 2019 Online (12-01-19), 4/120]



- (1) 10^{-4} s^{-1} (2) $4 \times 10^{-4} \text{ s}^{-1}$ (3) $2 \times 10^{-4} \text{ s}^{-1}$ (4) 10^{-6} s^{-1}



Answers

EXERCISE - 1

PART - I

- A-1.** (a) $4.5 \times 10^{-3} \text{ M sec}^{-1}$; (b) $3.0 \times 10^{-3} \text{ M sec}^{-1}$ **A-2.** (a) 40.5 g min^{-1} ; (b) 76.5 g min^{-1}
- A-3.** (a) $1.6 \times 10^{-2} \text{ atm min}^{-1}$; (b) $1.09 \times 10^{-5} \text{ mol liter}^{-1} \text{ sec}^{-1}$
- A-4.** (a) 3; (b) Both rates are $3.6 \times 10^{-3} \text{ mole dm}^{-3}\text{s}^{-1}$; (c) No effect; (d) Decreased by a factor of 8; No effect
- B-1.** (i) 9.994 M; (ii) $\frac{10}{2 \times (2 \times 10^{-6})}$; (iii) $5 \times 10^6 \text{ min}$ **B-2.** (a) $5 \times 10^{-5} \text{ M/s}$; (b) $4.2 \times 10^{-5} \text{ M/s}$
- B-3.** 0075. **B-4.** 1.73 time **C-1.** $1.25 \times 10^4 \text{ min}$; $2.5 \times 10^4 \text{ min}$
- C-2.** 39.2 minutes **C-3.** $1.84 \times 10^{-2} \text{ min}^{-1}$
- D-1.** (i) rate = $[A][B]$; (ii) $k = 4 \times 10^{-2} \text{ M}^{-1}\text{s}^{-1}$; (iii) rate = $2.8 \times 10^{-3} \text{ M}\cdot\text{s}^{-1}$
- D-2.** (a) $n = 2$; (b) First Order **D-3.** 2
- E-1.** (i) 2.484 hr^{-1} ; (ii) $\frac{50}{3} \text{ min.}$ **E-2.** $3.27 \times 10^{-3} \text{ min}^{-1}$
- E-3.** $0.000844 \text{ sec}^{-1}$ **F-1.** $k_1 = \frac{9 \ln 2}{2300}$; $k_2 = \frac{14}{2300} \ln 2$ **F-2.** $E_{\text{overall}} = E/3(2n+1)$
- G-1.** (i) 2, 1; (ii) $2.67 \times 10^8 \text{ mol}^{-2} \text{ litre}^2 \text{ sec}^{-1}$; (iii) $55.13 \text{ kJ mol}^{-1}$; (iv) $1.147 \times 10^{18} \text{ mol}^{-2} \text{ litre}^2 \text{ sec}^{-1}$
- G-2.** 80.65%. **G-3.** (A) $y + z$; (B) z ; (C) $x + y + z$; (D) x ; (E) $x + y$; (F) $-y$
- H-1.** - 1000 **H-2.** $k [\text{N}_2\text{O}_5]$ **H-3.** Rate = $k [\text{NO}]^2 [\text{Br}_2]$
- H-4.** (a) $2 \text{ B} + \text{F} \rightarrow 2 \text{ E}$; (b) A; (c) C, D; (d) rate = $k [\text{B}][\text{C}]$; (e) rate = $k' [\text{A}][\text{B}]$ (f) 2.
- I-1.** (30 days, 0.0231 per day) **I-2.** $(12.01 \times 10^{-5} \text{ years}^{-1}, 1/4)$
- I-3.** 0.9375 gram. **I-4.** $t = 4.5 \times 10^9 \text{ year.}$

PART - II

- A-1.** (A) **A-2.** (B) **A-3.** (D) **A-4.** (B) **A-5.** (B)
- A-6.** (C) **A-7.** (A) **A-8.** (D) **B-1.** (B) **B-2.** (C)
- B-3.** (C) **B-4.** (A) **B-5.** (A) **B-6.** (B) **B-7.** (D)
- B-8.** (B) **B-9.** (C) **C-1.** (D) **C-2.** (C) **C-3.** (D)
- C-4.** (C) **C-5.** (C) **D-1.** (B) **D-2.** (C) **D-3.** (B)
- D-4.** (B) **D-5.** (D) **D-6.** (C) **D-7.** (B) **D-8.** (B)



- | | | | | |
|-----------|----------|----------|----------|----------|
| E-1. (C) | E-2. (A) | E-3. (C) | E-4. (B) | E-5. (B) |
| F-1. (B) | F-2. (D) | F-3. (D) | F-4. (B) | G-1. (B) |
| G-2. (D) | G-3. (B) | G-4. (C) | G-5. (B) | H-1. (D) |
| H-2. (A) | H-3. (C) | H-4. (C) | H-5. (C) | H-6. (B) |
| H-7. (A) | I-1. (D) | I-2. (A) | I-3. (B) | I-4. (C) |
| I-5. (B) | I-6. (C) | I-7. (A) | I-8. (A) | I-9. (D) |
| I-10. (D) | | | | |

PART - III

- $(A \rightarrow r) ; (B \rightarrow s) ; (C \rightarrow q) ; (D \rightarrow p)$
- (A) p, q, r, s ; (B) q, r, s ; (C) p, q, r, s ; (D) p, r, s
- (A – r ; B – s ; C – p ; D – q)

EXERCISE - 2

PART - I

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (C) | 2. (C) | 3. (A) | 4. (B) | 5. (B) |
| 6. (C) | 7. (A) | 8. (C) | 9. (B) | 10. (B) |
| 11. (C) | 12. (A) | 13. (B) | 14. (A) | 15. (A) |
| 16. (D) | 17. (A) | 18. (C) | 19. (A) | 20. (B) |

PART - II

- | | | | | |
|-----------------------|---------|------|-------|------|
| 1. 1 | 2. Zero | 3. 2 | 4. 54 | 5. 3 |
| 6. 50 | 7. 2 | 8. 2 | 9. 2 | |
| 10. 10^{-19} minute | 11. 30 | | | |

PART - III

- | | | | | |
|-----------|----------|----------|------------|-----------|
| 1. (ABD) | 2. (BCD) | 3. (ABC) | 4. (ABD) | 5. (ABCD) |
| 6. (AD) | 7. (AC) | 8. (BC) | 9. (ABC) | 10. (AB) |
| 11. (ABC) | 12. (CD) | 13. (BC) | 14. (ABCD) | 15. (BC) |
| 16. (AB) | | | | |

PART - IV

- | | | | | |
|---------|---------|---------|--------|---------|
| 1. (A) | 2. (D) | 3. (B) | 4. (A) | 5. (C) |
| 6. (A) | 7. (C) | 8. (B) | 9. (B) | 10. (A) |
| 11. (C) | 12. (A) | 13. (C) | | |





EXERCISE - 3

PART - I

- | | | |
|--|----------------------------------|-----------------------|
| 1. (D) | 2. $t = 4.62 \times 10^9$ years. | 3. $E_A = 100$ KJ/mol |
| 4. (B) | 5. $t_{1/2} = 24$ min. | 6. (A) |
| 7. $(t_{1/2})_1 = 33.68 \text{ hr}^{-1}$, $(t_{1/2})_2 = 67.37 \text{ hr}^{-1}$, $(t_{1/2})_3 = 29.77 \text{ hr}^{-1}$. | 8. (B) | 9. (A) |
| 10. (a) $R_0 = k[A_0]$, (b) 0.5 sec^{-1} . | 11. 950 mm of Hg | 12. (C) |
| 14. (A) | 15. (D) | 16. (A) |
| 17. (D) | 18. 0 | 19. (A) |
| 20.* (ABD) | 21. 9 | 22. (D) |
| 23. (A) | 24. (B) | 25. (B) |
| 26. 8 | 27. (BCD) | 28.* (AB) |
| 29.* (AD) | 30. 8500 J/mole | |

PART - II

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (3) | 2. (4) | 3. (2) | 4. (2) | 5. (4) |
| 6. (2) | 7. (1) | 8. (3) | 9. (2) | 10. (4) |
| 11. (4) | 12. (3) | 13. (2) | 14. (3) | 15. (4) |
| 16. (2) | 17. (2) | 18. (4) | 19. (4) | 20. (2) |
| 21. (1) | 22. (4) | 23. (1) | | |